Pakistan Ordnance Factory (POF) Wah Cantt



Department of IT (CO – 4)

Internship Final Report

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Submission Date : 14 September 2026 (Monday)

Declaration

I hereby declare that this internship report presents the technical learning, practical activities, observations, and troubleshooting experience gained during my six-week internship in the field of Database Administration. The report has been prepared for academic and organizational submission and summarizes the major activities performed during the internship.

Student Signature: _____

Date: _____

Acknowledgement

I am thankful to Almighty Allah for providing me with the opportunity, strength, and guidance to complete my internship successfully. I would like to express my sincere gratitude to my university, faculty members, internship supervisor and the professionals who guided me throughout the training.

Special appreciation is extended to the instructor and technical staff for providing practical guidance in Oracle Database Administration, Oracle Linux, SQL Developer, TOAD, database monitoring, backup concepts, connectivity troubleshooting and real-world Data Center operations.

This internship provided valuable exposure to both theoretical concepts and practical DBA activities. The support and guidance received during the internship helped me develop a better understanding of database administration and professional IT infrastructure.

Abstract

This report presents the work completed during a six-week internship focused on Database Administration and Oracle Database technologies. The internship provided practical exposure to Oracle Linux, VMware, Oracle Database 12c, SQL Developer, TOAD, database design, database administration, monitoring, backup concepts and troubleshooting.

During Week 1, VMware and Oracle Linux were prepared as the base environment for database administration. Week 2 focused on installing Oracle Database 12c on Oracle Linux. Week 3 introduced SQL Developer and TOAD and established connections with Oracle Database 12c. During Week 4, a Boutique Management System was designed and developed in SQL Developer, including the basic database structure and sample data. Week 5 focused on database administration through TOAD, including remote database access, backup concepts and Oracle Enterprise Manager (OEM) monitoring. Week 6 concluded the internship with a Data Center visit and a summary of challenges and errors encountered during the training.

The internship improved practical understanding of database installation, configuration, connectivity, security, monitoring, administration and troubleshooting. It also provided an opportunity to relate academic database concepts to real-world infrastructure and professional DBA responsibilities.

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1. Introduction

1.1 Internship Overview

The internship was completed as a practical training program in Database Administration (DBA). The main focus was Oracle Database 12c and the supporting tools and operating system environment required for database administration.

1.2 Purpose of the Internship

The internship was designed to provide practical exposure to database installation, configuration, connectivity, administration, monitoring, security, backup concepts, database development, and troubleshooting.

1.3 Scope of the Training

- VMware and Oracle Linux environment preparation
- Oracle Database 12c installation and configuration
- SQL Developer and TOAD connectivity
- Database design and Boutique Management System project
- Database administration and remote access
- Backup concepts and database health monitoring
- Oracle Enterprise Manager introduction
- Data Center observation and DBA troubleshooting

1.4 Report Organization

This report summarizes the six-week internship chronologically. Unlike the detailed weekly practical reports, this final organizational report focuses on the main activities, learning outcomes, tools, and professional skills developed during the internship.

2. Internship Organization and Working Environment

2.1 Host Organization

The internship was conducted in a professional technical environment where database administration concepts were demonstrated through practical Oracle Database activities. The environment included Oracle Linux virtual machines, Oracle Database 12c, Windows-based database client tools, and a real Data Center visit.

2.2 Technical Environment

Component	Environment / Tool
Virtualization	VMware Workstation
Operating System	Oracle Linux 7.x
Database	Oracle Database 12c Enterprise Edition
Database Client	SQL Developer
Administration Tool	TOAD
Monitoring Tool	Oracle Enterprise Manager / EM Express
Database Connectivity	Oracle Listener, TNS, TCP/IP
Project	Boutique Management System

2.3 Working Approach

The training followed a practical approach. Concepts were first introduced by the instructor and then implemented in a controlled laboratory environment. Errors were analyzed using Oracle error messages, Linux commands, listener utilities, and database administration queries.

2.4 Role of DBA

The training emphasized the DBA's responsibility for database availability, security, performance, users, privileges, storage, connectivity, backup, monitoring, and troubleshooting.

3. Internship Objectives

- Understand the role and responsibilities of a Database Administrator.
- Prepare a virtual Oracle Linux environment using VMware.
- Install and configure Oracle Database 12c.
- Connect database client tools with Oracle Database.
- Understand database design and develop a small relational project.
- Perform basic database administration using TOAD.
- Understand remote database access and connectivity.
- Understand backup types and their administrative purpose.
- Monitor database health using Oracle Enterprise Manager.
- Observe a professional Data Center environment.
- Develop practical troubleshooting and problem-solving skills.

3.1 Expected Outcomes

By the end of the internship, the trainee was expected to understand the complete basic workflow of an Oracle DBA: preparing the server environment, installing the database, establishing client connectivity, creating and managing database objects, performing administration tasks, monitoring database health, and troubleshooting common errors.

4. Week 1 : VMware and Oracle Linux Setup

4.1 Overview

The first week established the foundation for the complete Oracle Database laboratory. VMware was used to create a virtual machine and Oracle Linux was attached and configured as the operating system for the Oracle Database environment.

4.2 VMware Installation

VMware Workstation was installed on the host computer. A virtual machine was created with appropriate memory, processor, disk, and network settings. Virtualization provided an isolated environment in which Oracle Linux and Oracle Database could be installed without changing the main operating system.

4.3 Oracle Linux Installation

Oracle Linux was installed in the VMware virtual machine. Basic operating system configuration was completed so that the environment could later support Oracle Database 12c.

4.4 Basic Linux Preparation

The Linux environment was checked for network connectivity, user access, permissions, storage directories, and required packages. These steps prepared the virtual machine for Oracle Database installation.

4.5 Main Learning

The major learning outcome of Week 1 was understanding how virtualization can be used to build a controlled database server environment and how Oracle Linux provides the platform for Oracle Database.

Figure 1 : VMware Workstation Home Screen

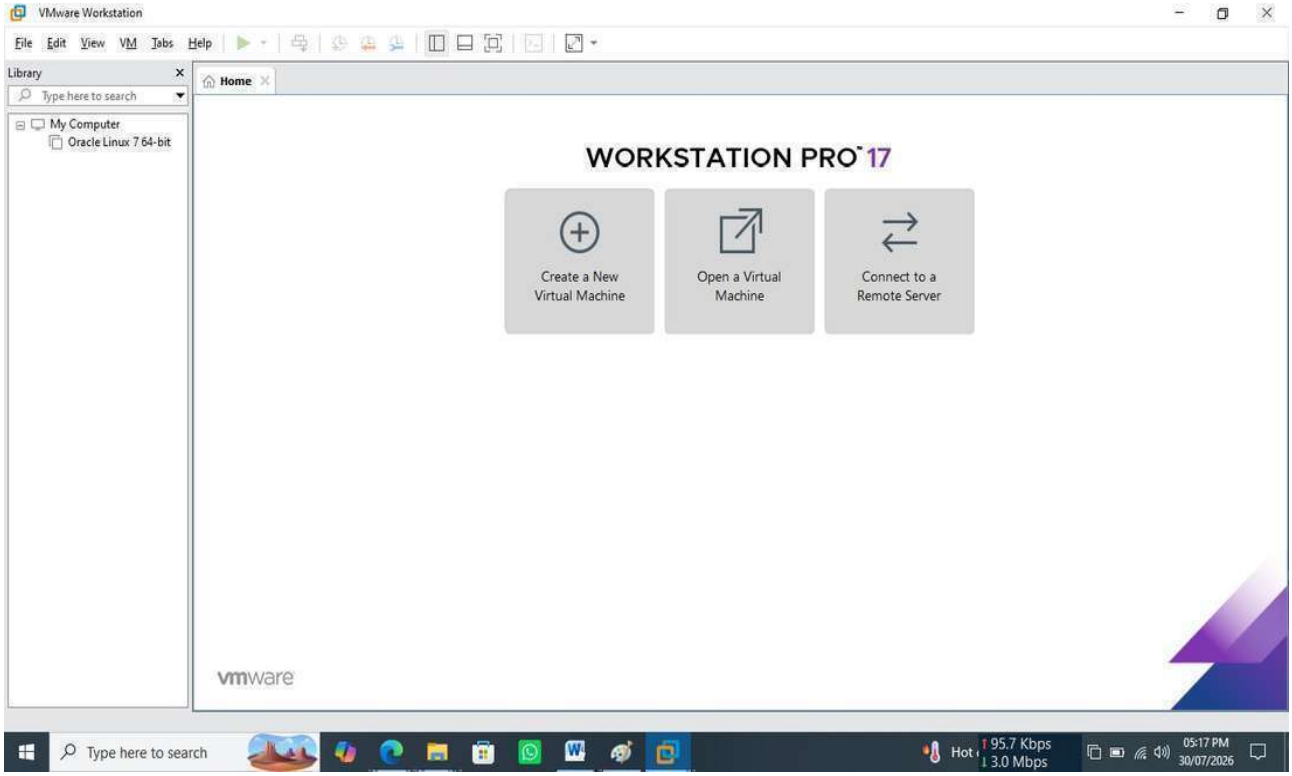


Figure 2 : Oracle Linux Installation Screen

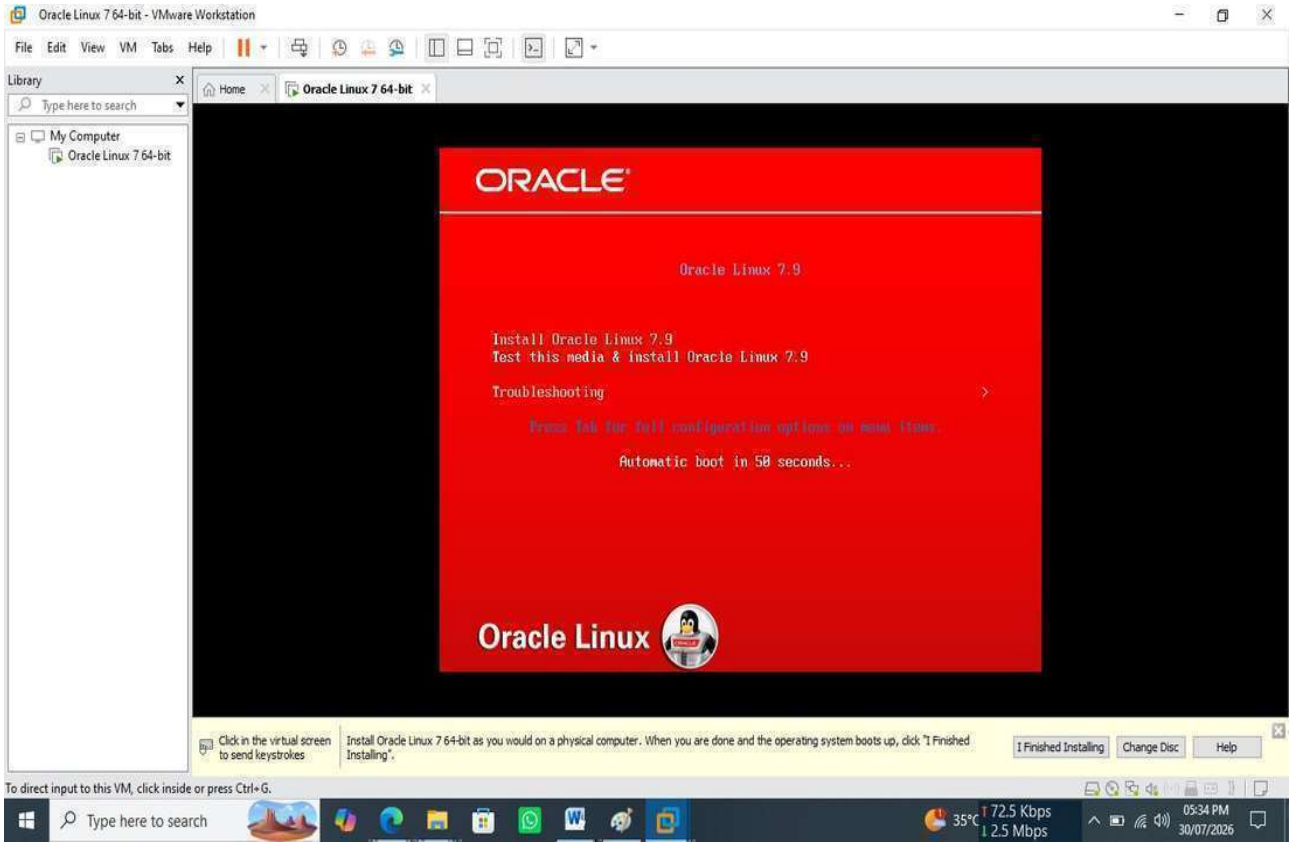


Figure 3 : Oracle Linux Login Screen

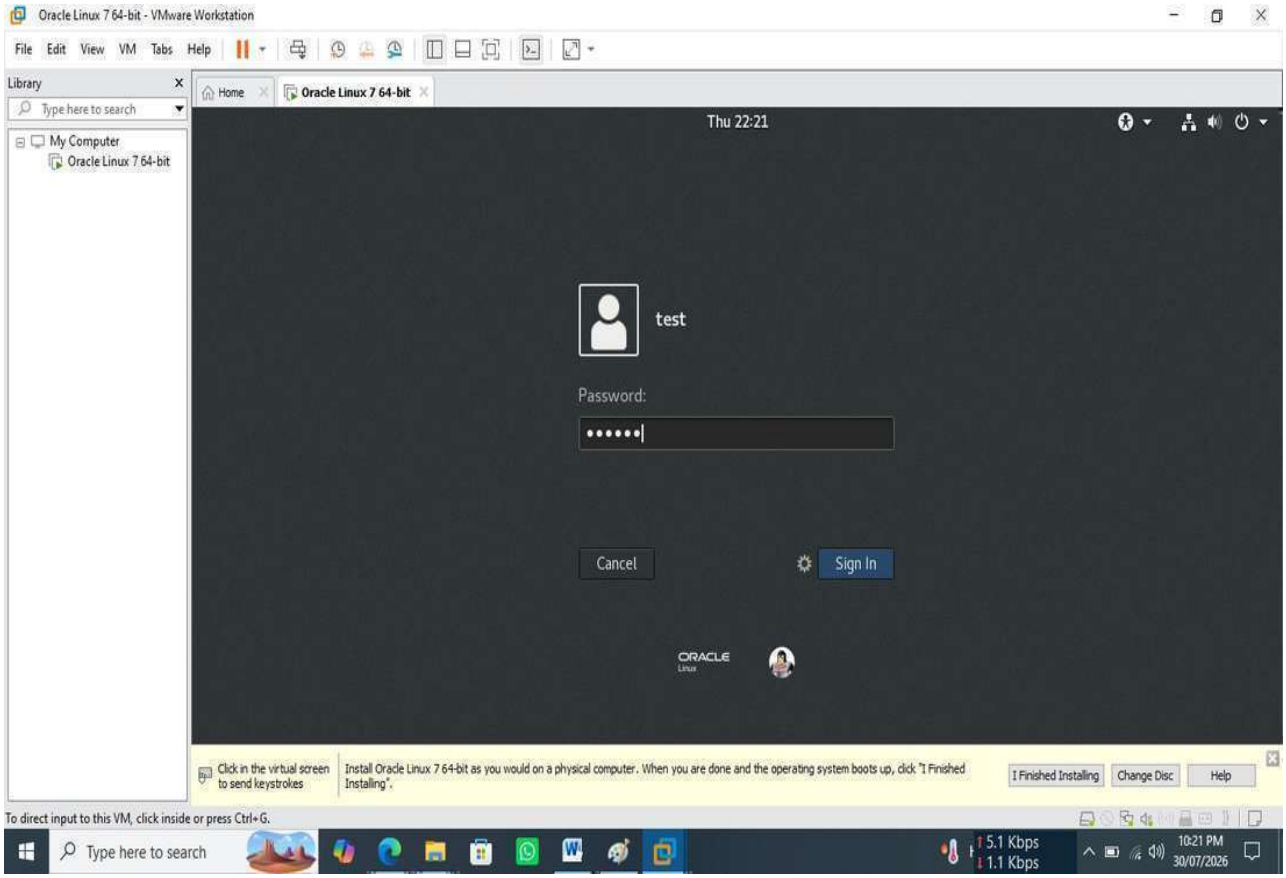
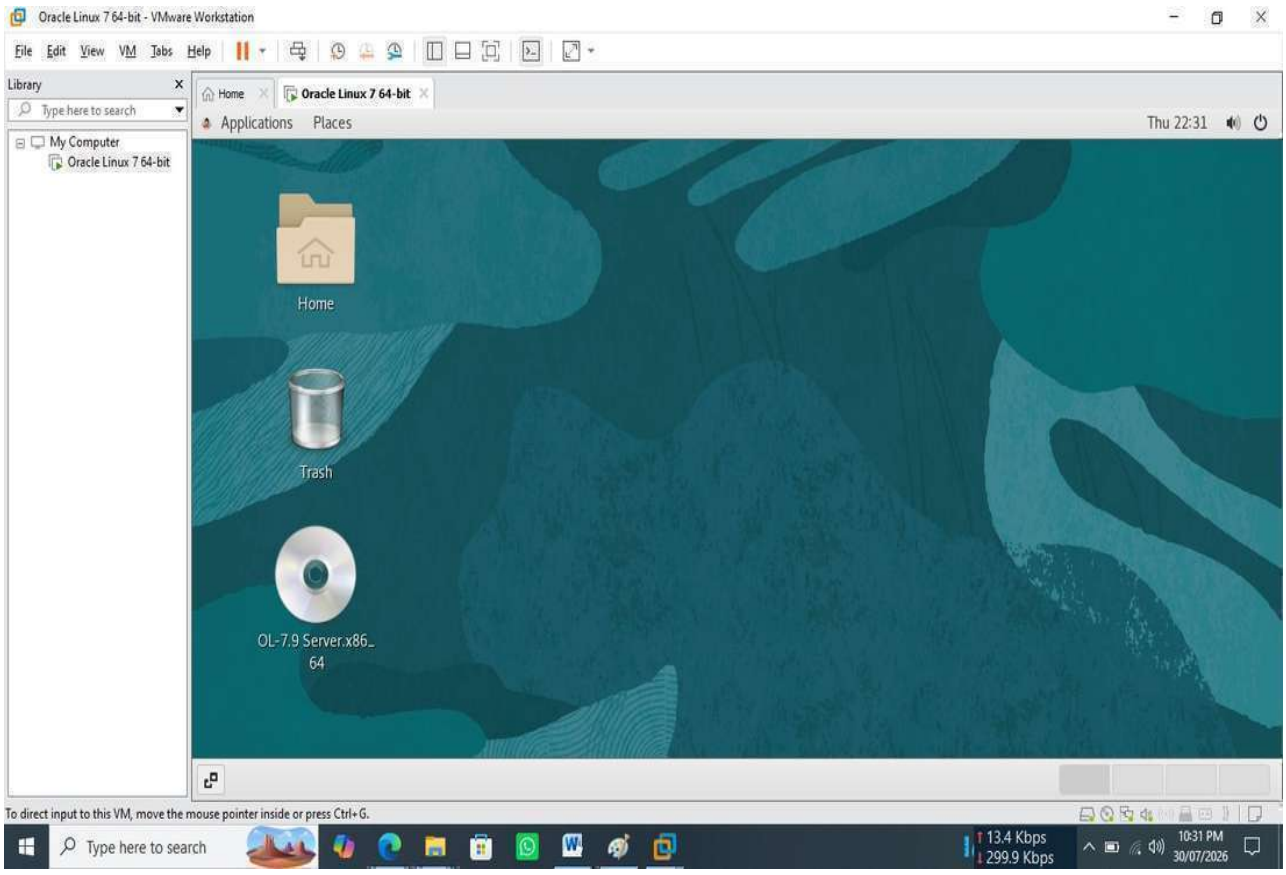


Figure 4 : Oracle Linux Desktop Environment



5. Week 2 : Oracle Database 12c Installation on Oracle Linux

5.1 Overview

Week 2 focused on the installation and initial configuration of Oracle Database 12c on the Oracle Linux virtual machine.

5.2 Installation Preparation

The Oracle Linux environment was prepared with the required directories, Oracle user, groups, permissions, packages, and installation files. The Oracle Database installation media was made available to the virtual machine.

5.3 Oracle Database 12c Installation

Oracle Database 12c was installed on Oracle Linux using the Oracle installation process. The Oracle home and database software components were configured according to the laboratory environment.

5.4 Database Configuration

After installation, the database instance and supporting services were checked. Oracle environment variables such as ORACLE_HOME and ORACLE_SID were configured so that Oracle utilities could be executed from the Linux terminal.

5.5 Listener and Database Startup

The Oracle Listener was started and database status was checked. Basic commands such as lsnrctl status and SQL*Plus were used to verify that the Oracle environment was operational.

5.6 Main Learning

Week 2 provided practical understanding of Oracle Database installation, Linux-based configuration, Oracle environment variables, database startup, and listener management.

Figure 1 : Oracle Universal Installer

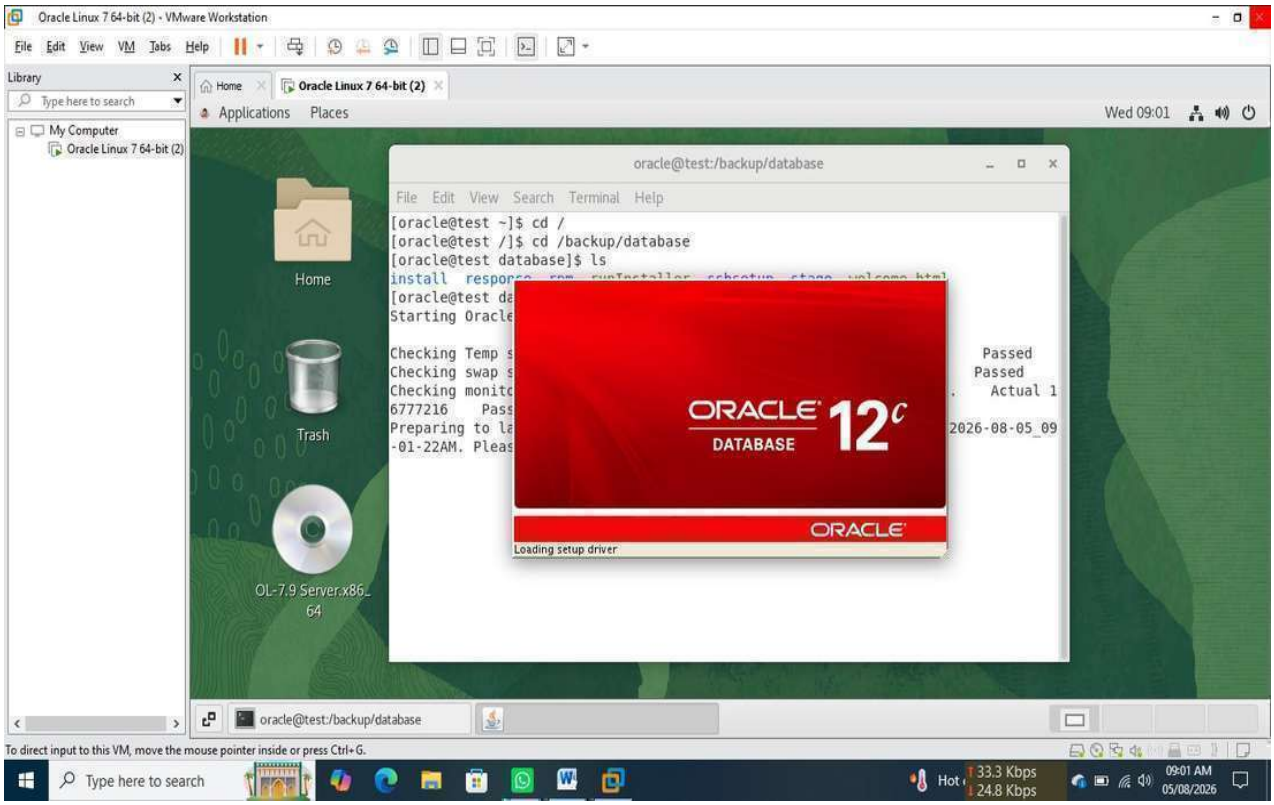


Figure 2 : Oracle Database Installation Progress

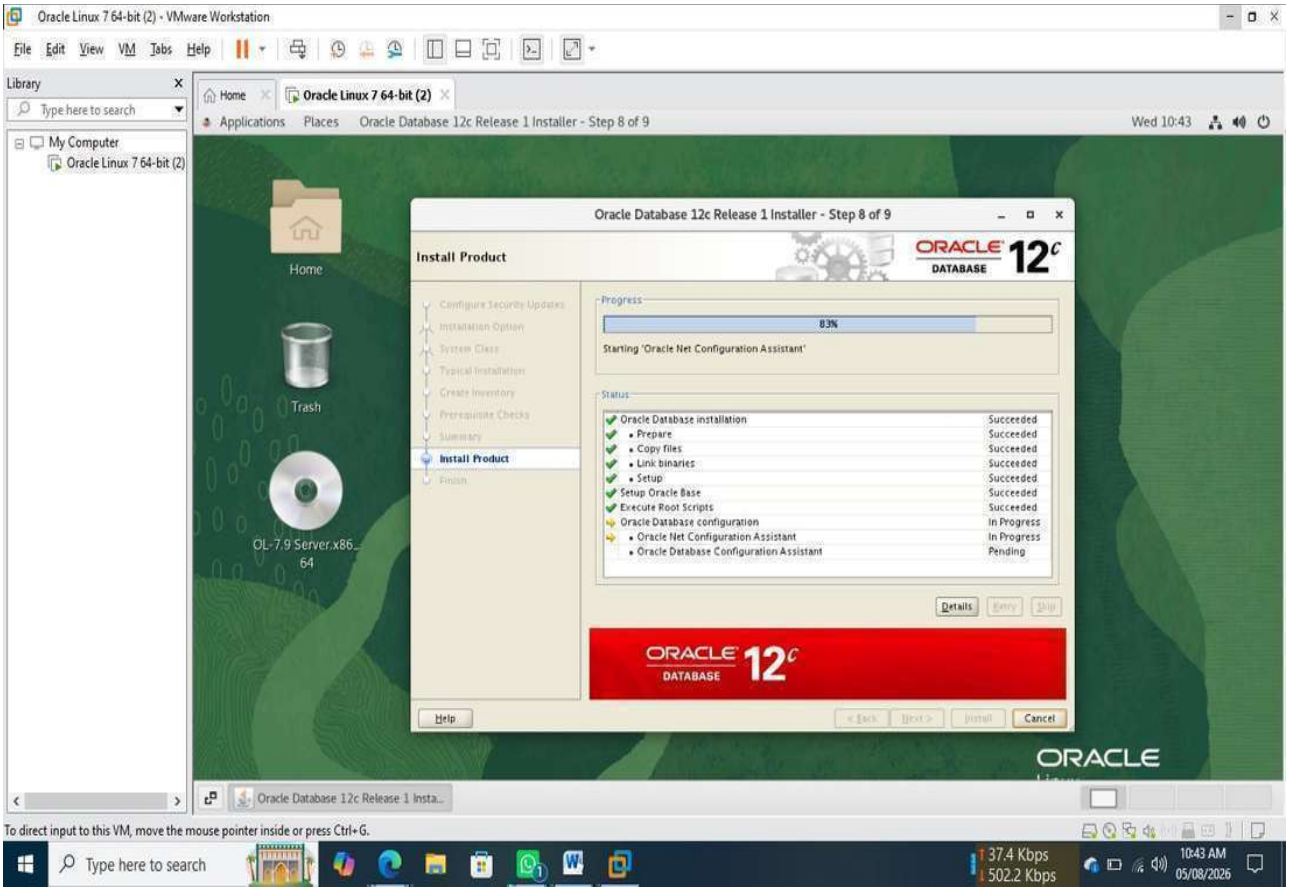


Figure 3 : Oracle Environment Variables

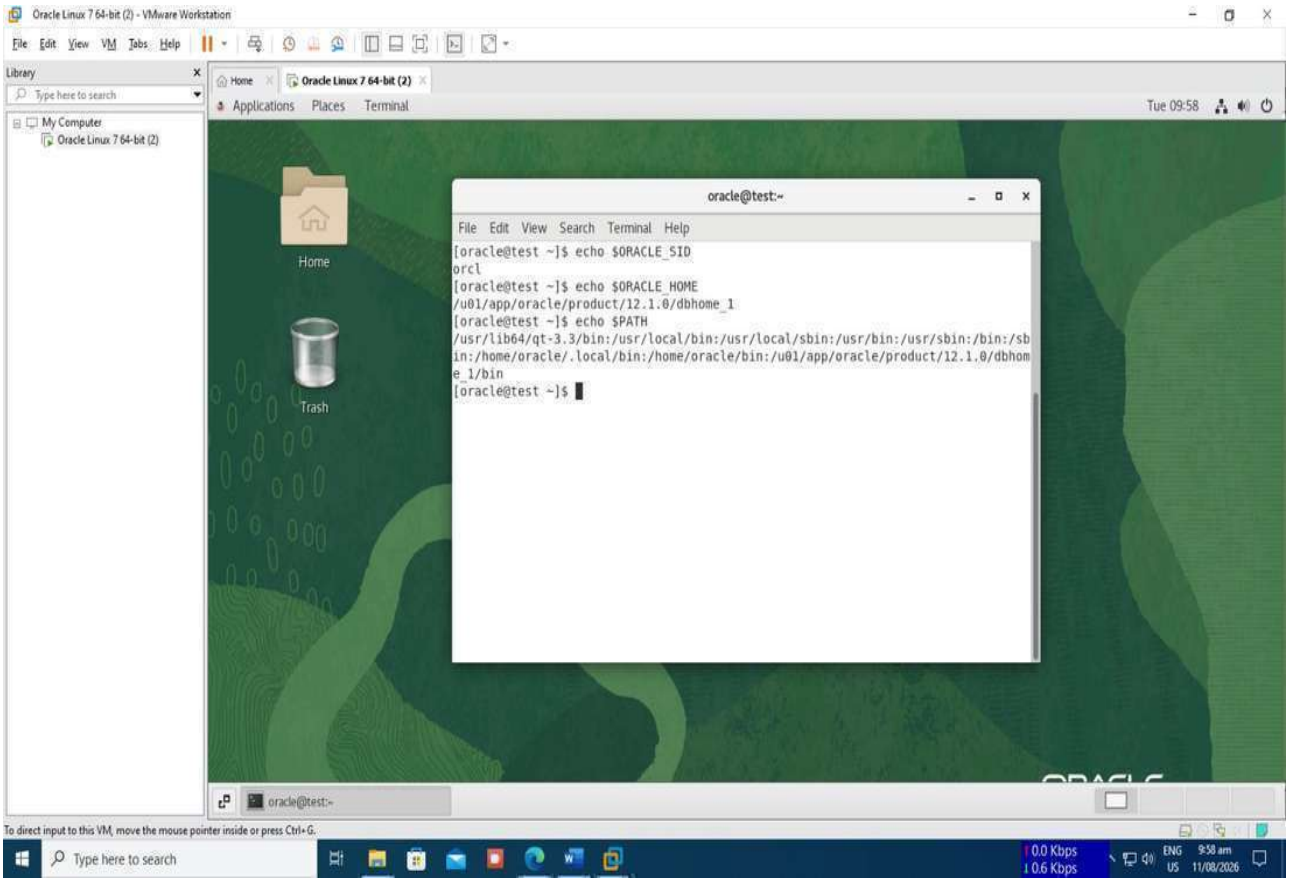


Figure 4 : Starting Oracle Net Listener

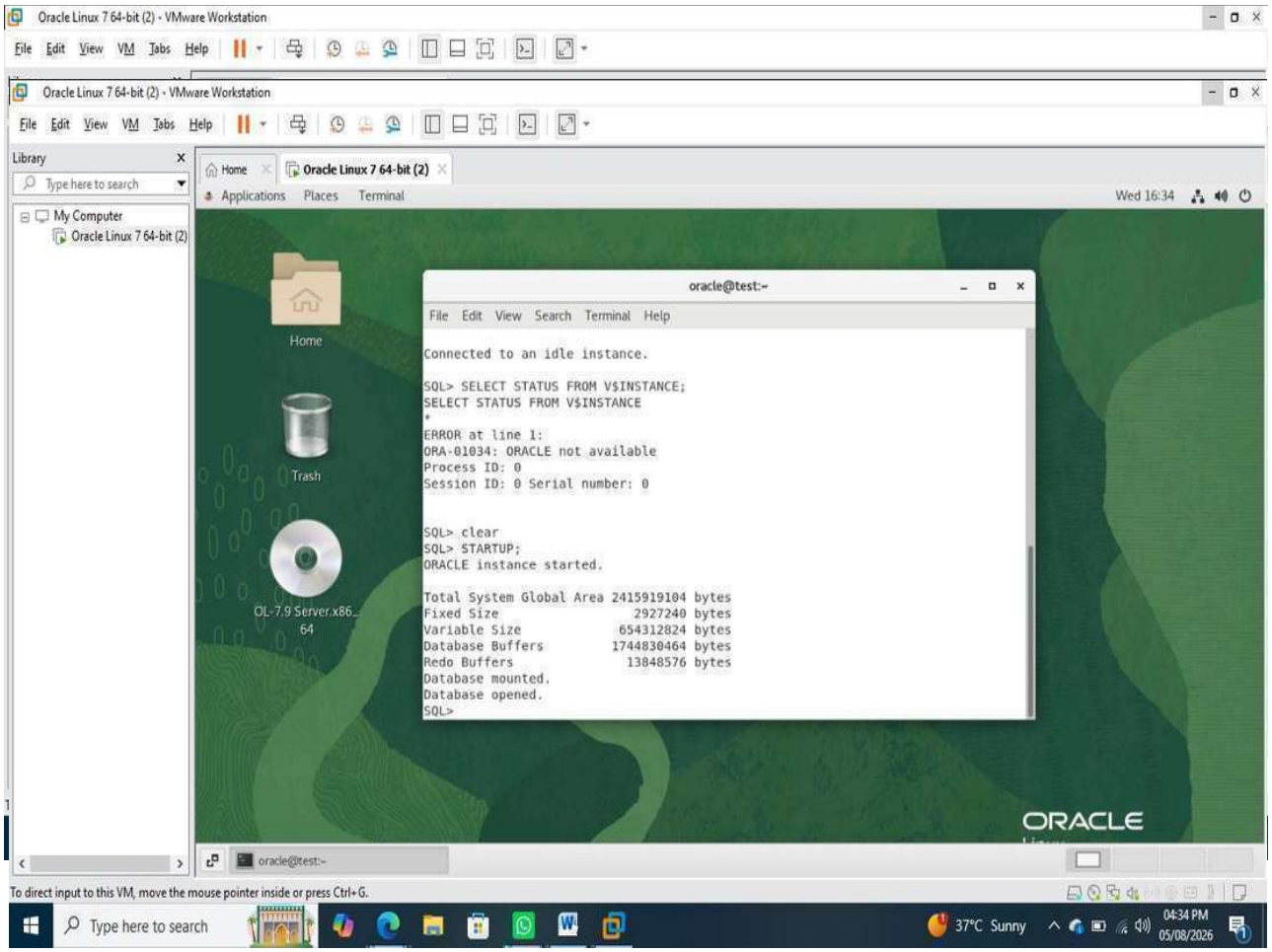
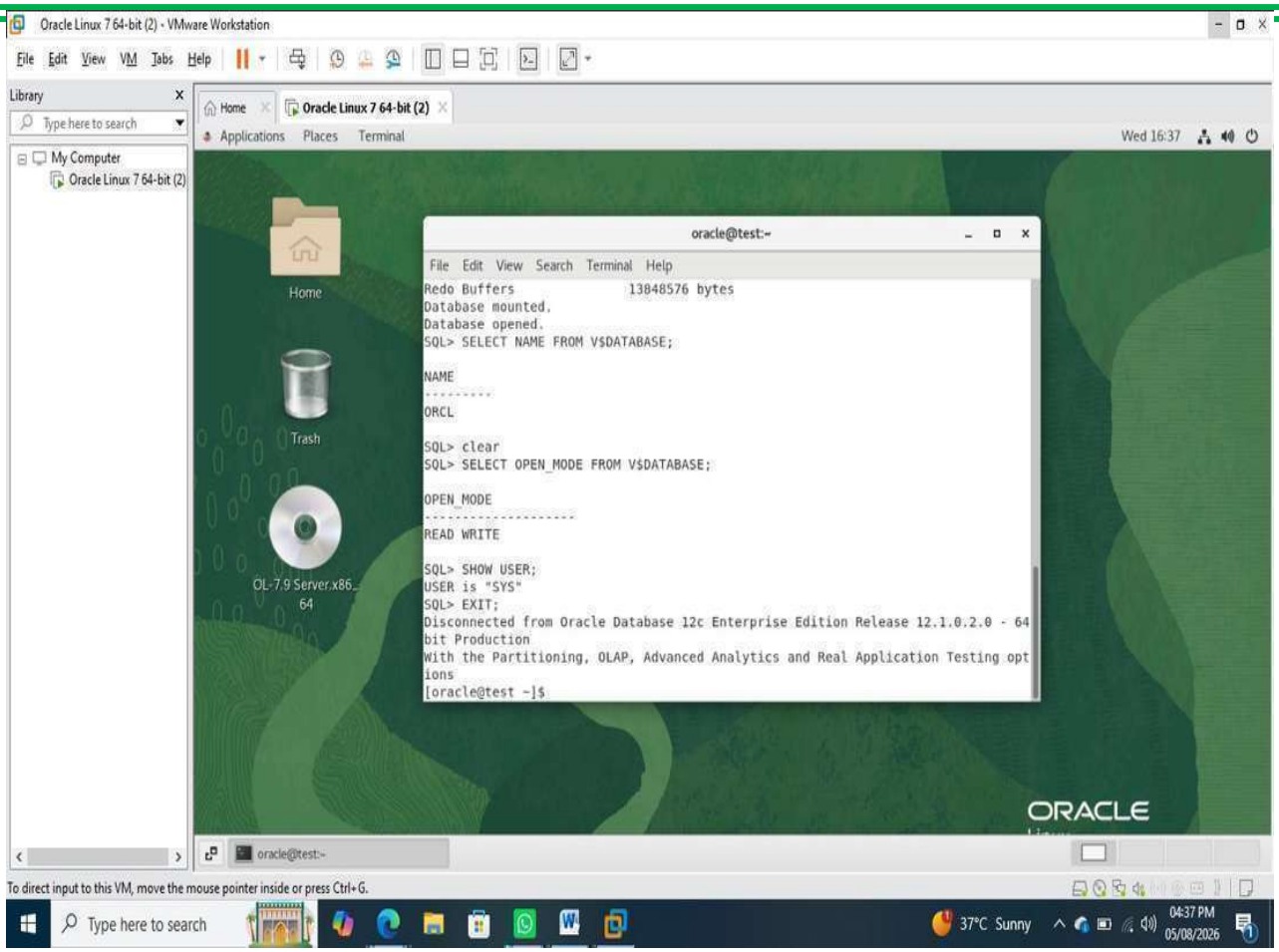


Figure 5 : Starting Oracle Database

Figure 6 : Exiting SQL*Plus



6. Week 3 : SQL Developer and TOAD Connectivity

6.1 Overview

Week 3 focused on connecting graphical database client tools with Oracle Database 12c. SQL Developer and TOAD were introduced as tools for database development and administration.

6.2 SQL Developer Connection

SQL Developer was installed and configured on the Windows system. A database connection was created using the Oracle Database server address, listener port, service name, and valid credentials.

6.3 TOAD Connection

TOAD was introduced as a professional Oracle administration and development tool. A connection was configured to access Oracle Database 12c and perform database-related tasks.

6.4 Connectivity Verification

The connection process demonstrated the importance of the Oracle Listener, service registration, TNS configuration, host/IP address, port 1521, and database credentials.

6.5 Main Learning

The main learning of Week 3 was understanding how client applications communicate with Oracle Database and how a DBA can troubleshoot connection problems.

Figure 1 : Launching Oracle SQL Developer

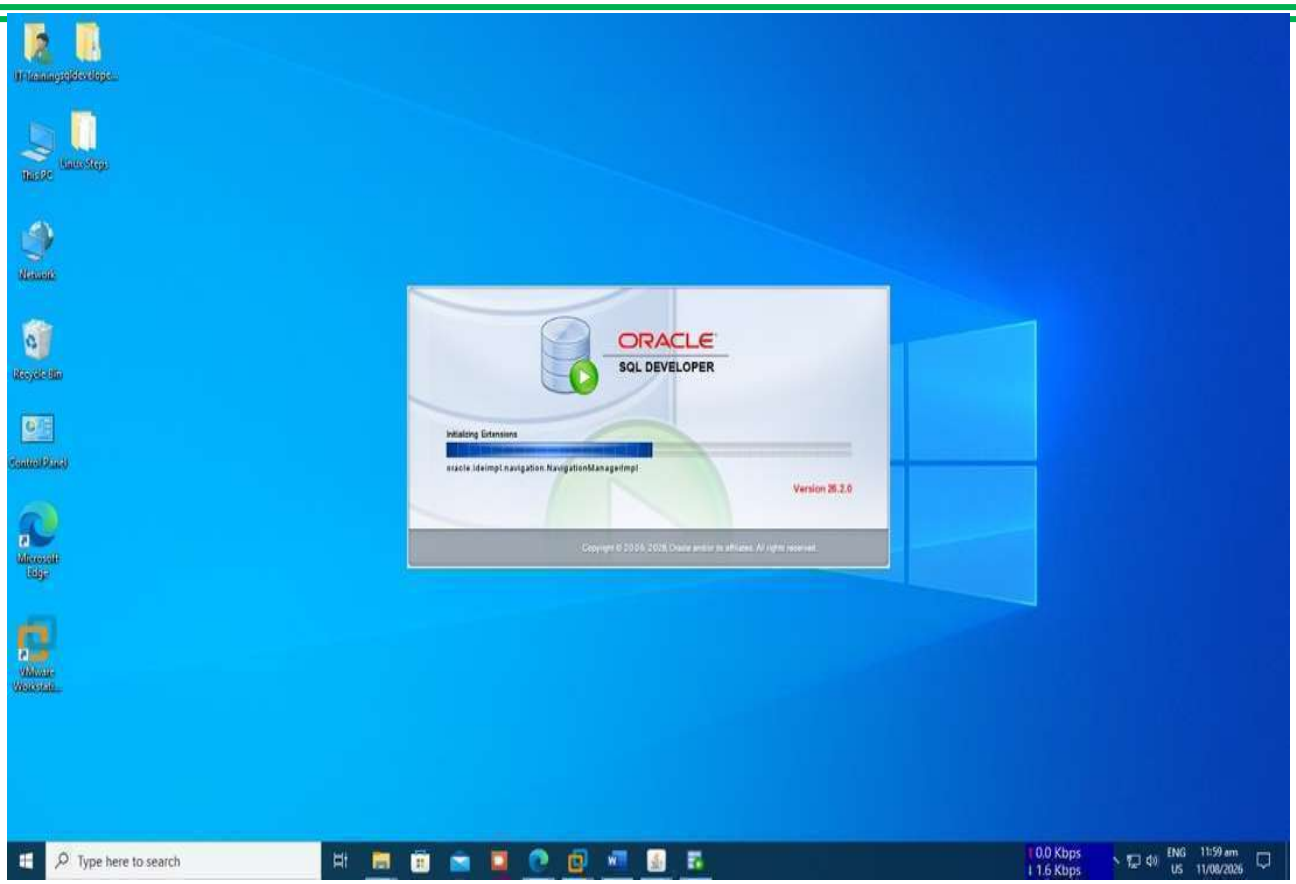


Figure 2 : Oracle SQL Developer Home Screen

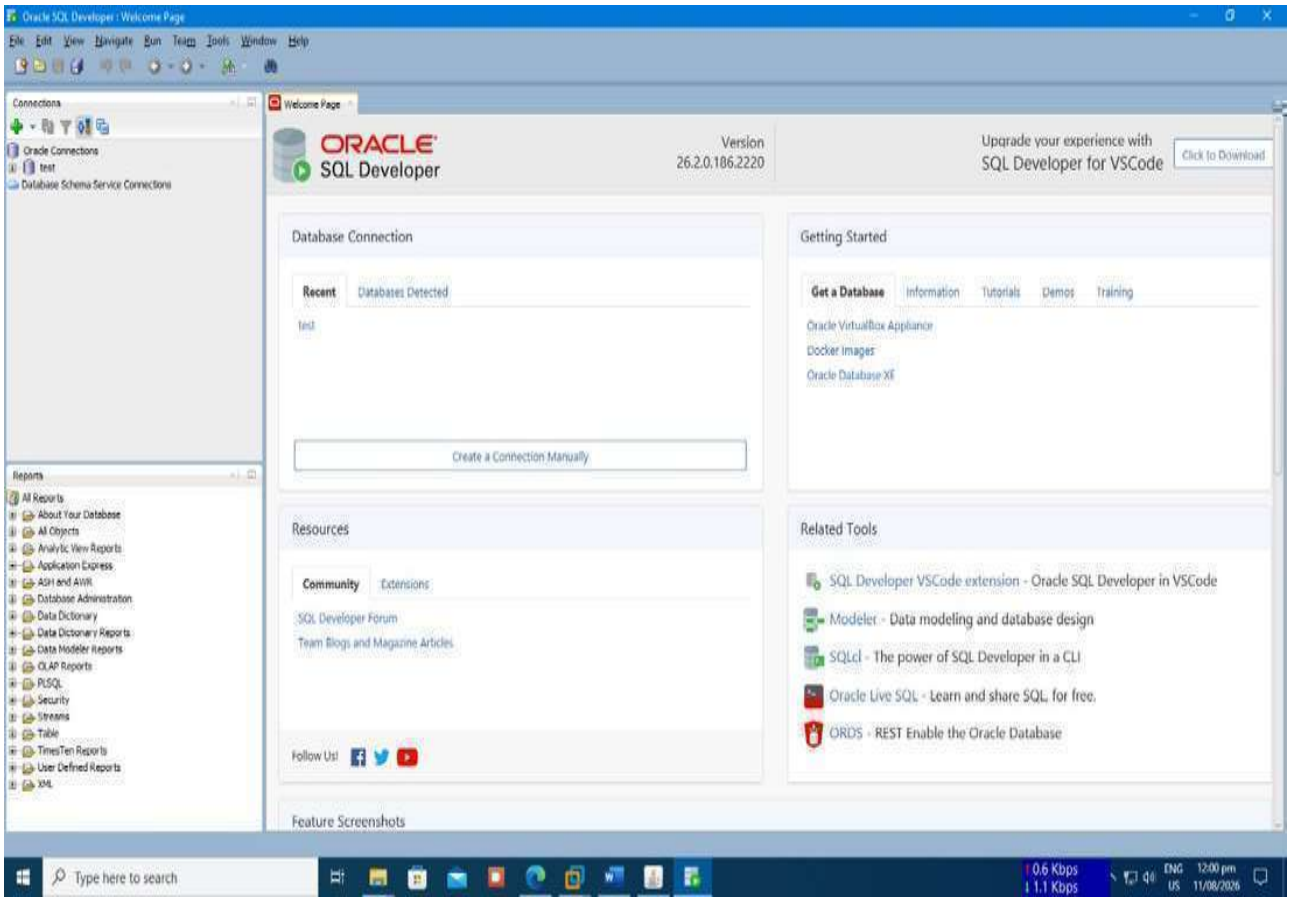


Figure 3 : Database Connection Configuration

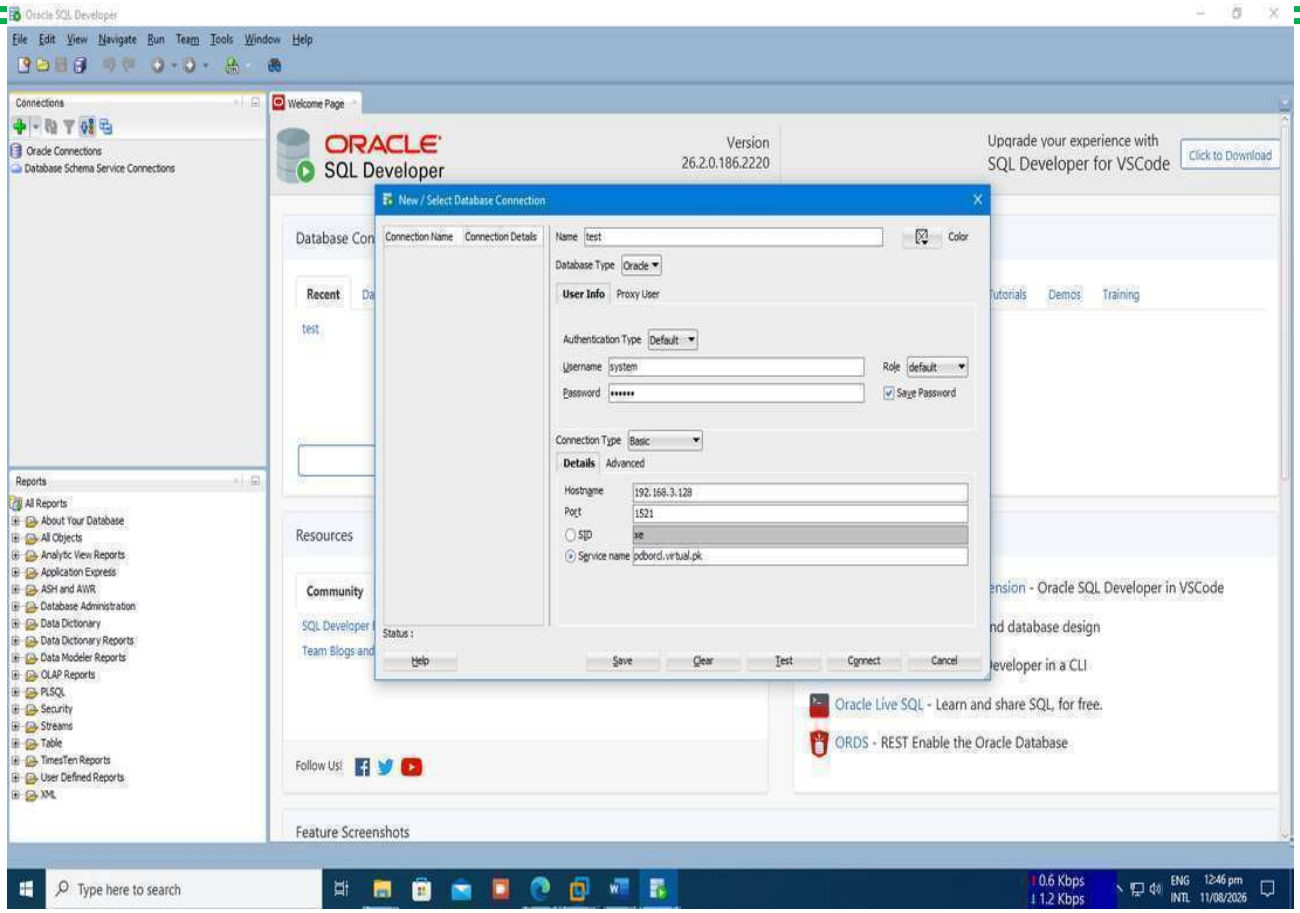


Figure 4 : SQL Worksheet

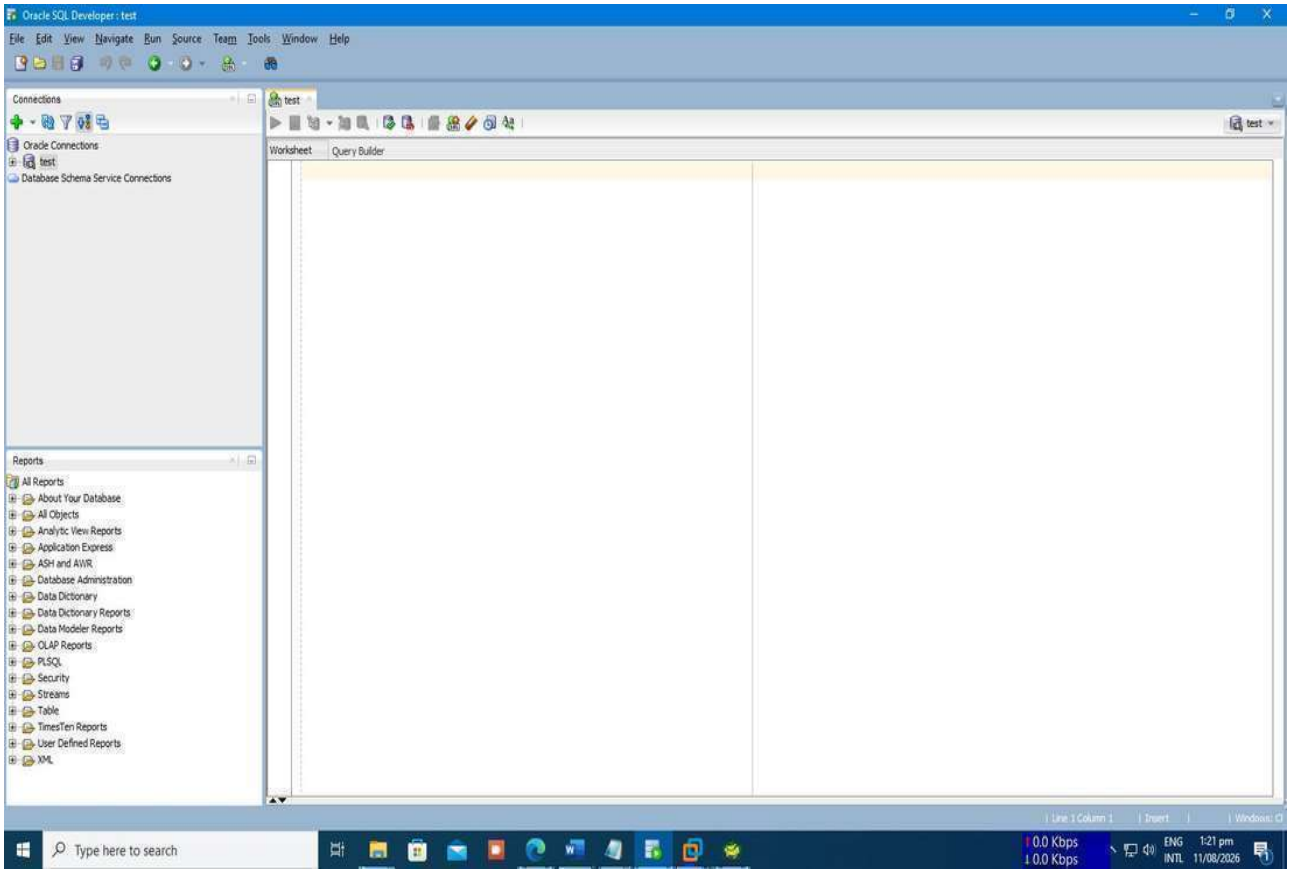


Figure 5 : Toad for Oracle Download Page

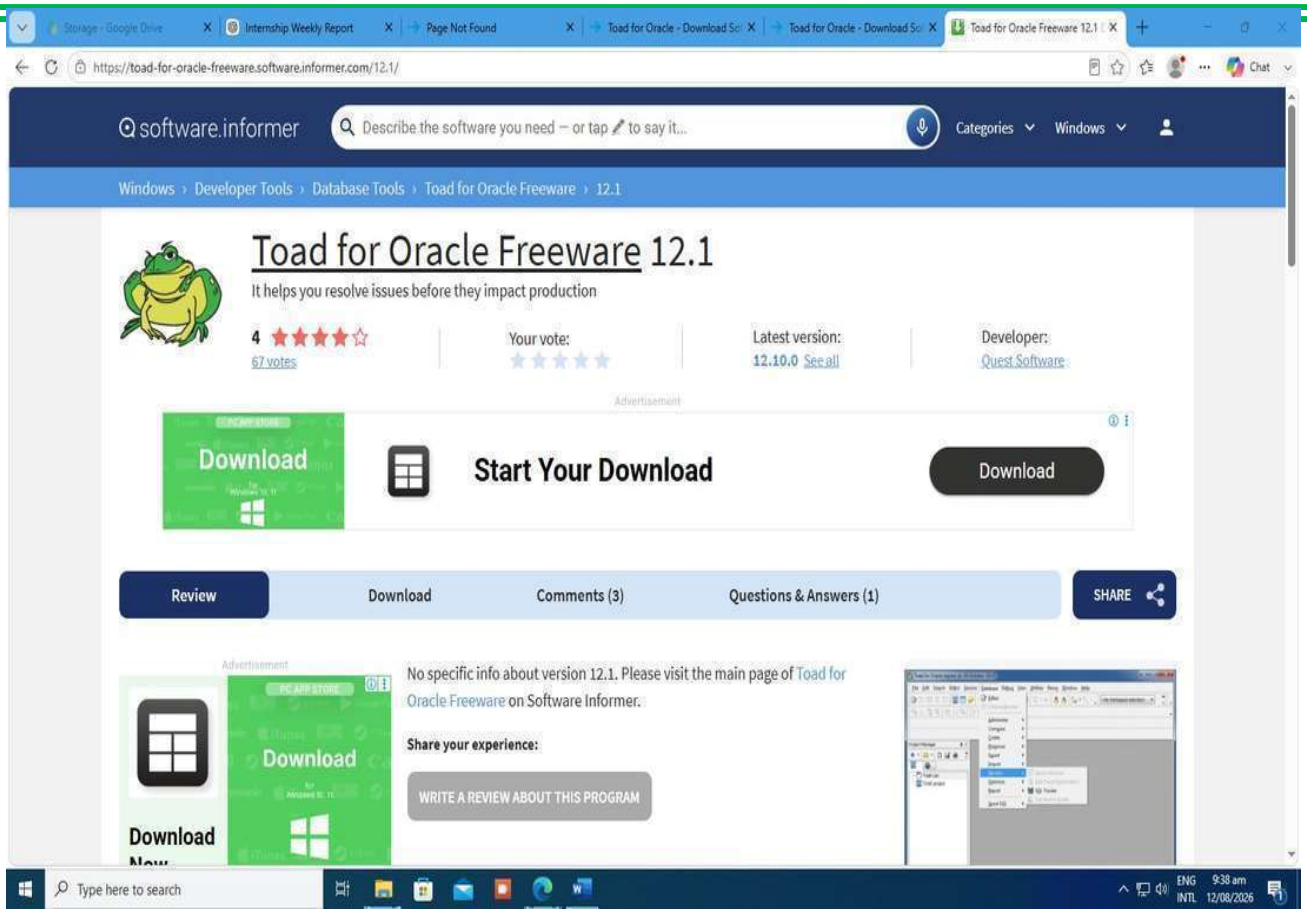


Figure 6 : Toad Installation in Progress

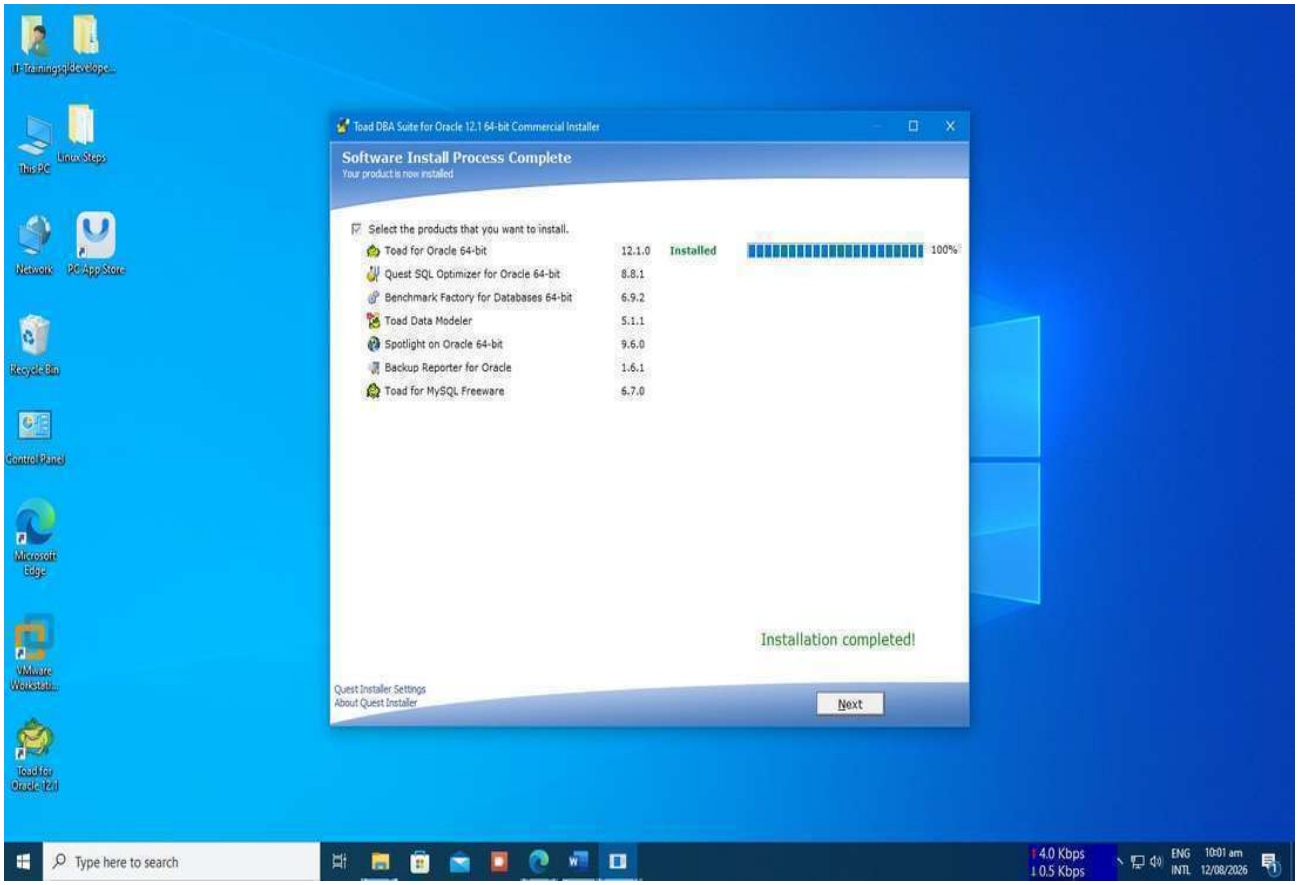


Figure 7 : Toad Oracle Connection Settings

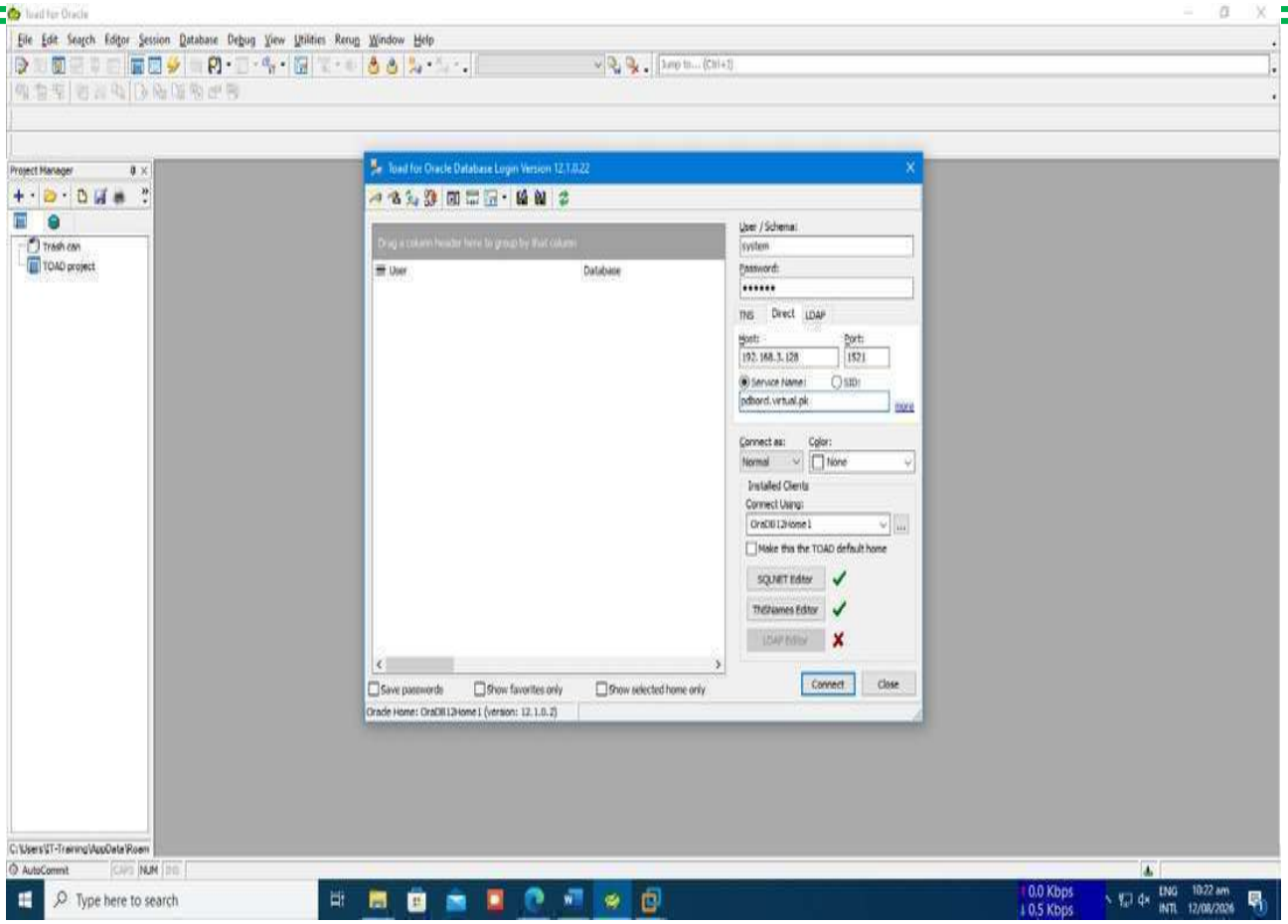


Figure 8 : Entering License Key and Site Message In Toad Application

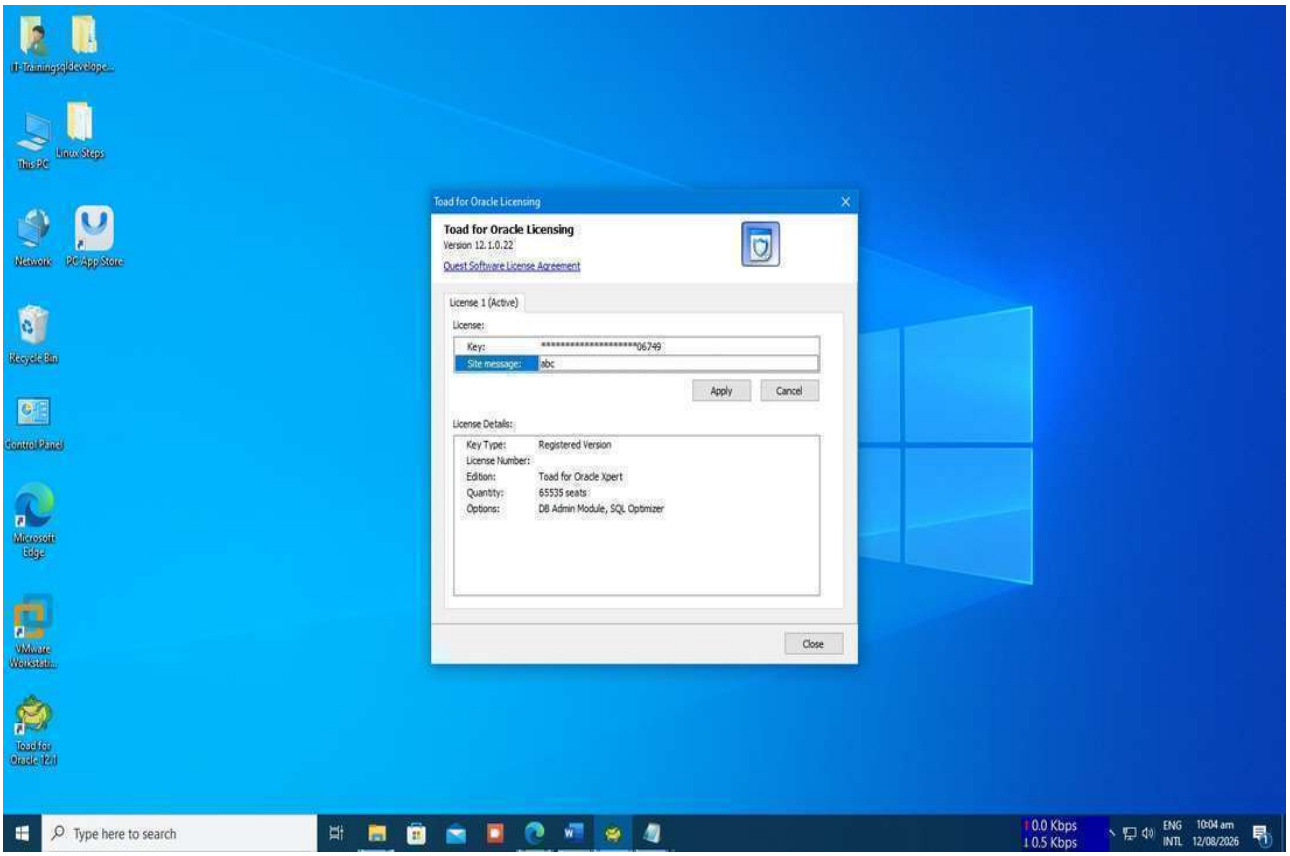
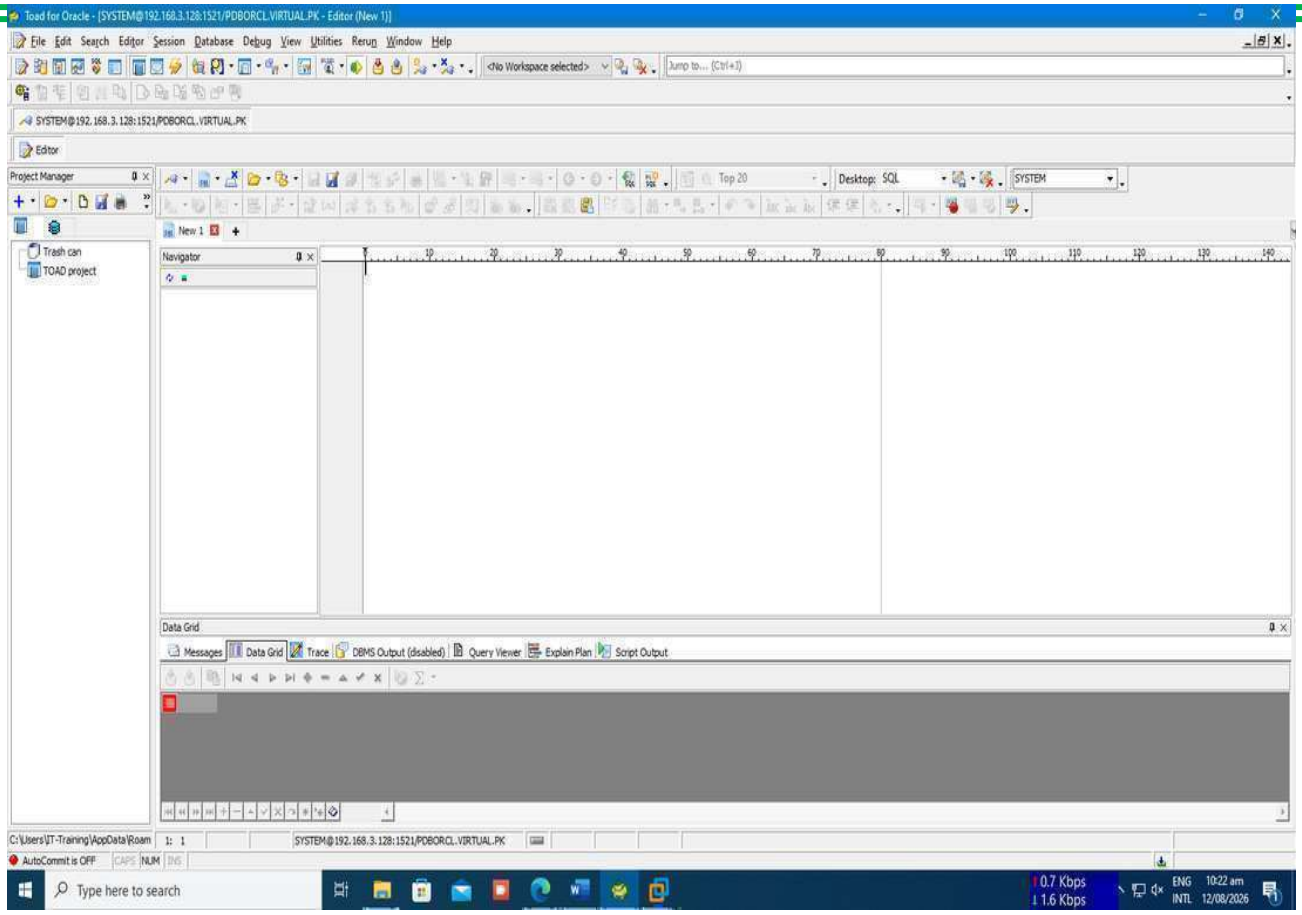


Figure 9 : Toad Connected to Oracle Database



7. Week 4 : Boutique Management System Database Project

7.1 Project Overview

During Week 4, a database project titled Boutique Management System was developed in SQL Developer. The purpose of the project was to apply relational database concepts in a practical application.

7.2 Project Objectives

- Design a relational database for boutique management.
- Identify major entities and their relationships.
- Create tables and define appropriate constraints.
- Insert sample records for testing.
- Prepare the database for application queries and future forms/reports.

7.3 Database Design

The database design was prepared by identifying the main business entities and their relationships. The design considered customers, products, employees, orders, order details, payments, and related information.

7.4 ERD — Entity Relationship Diagram

The Entity Relationship Diagram (ERD) represents the main entities of the Boutique Management System and the relationships among them. The ERD should be inserted below as the main visual representation of the database design.

7.5 Tables and Constraints

The database tables were created according to the project design. Primary keys, foreign keys, and other appropriate constraints were considered to maintain data integrity.

7.6 Sample Data

Sample records were inserted into the project tables to test the database structure and demonstrate basic data operations. Detailed creation and insertion commands are intentionally omitted from this final report because they are already documented in the weekly practical report.

7.7 Main Learning

Week 4 strengthened understanding of database design, relational modeling, constraints, sample data, and the relationship between business requirements and database structures.

Figure 1 : All Boutique Management System Tables

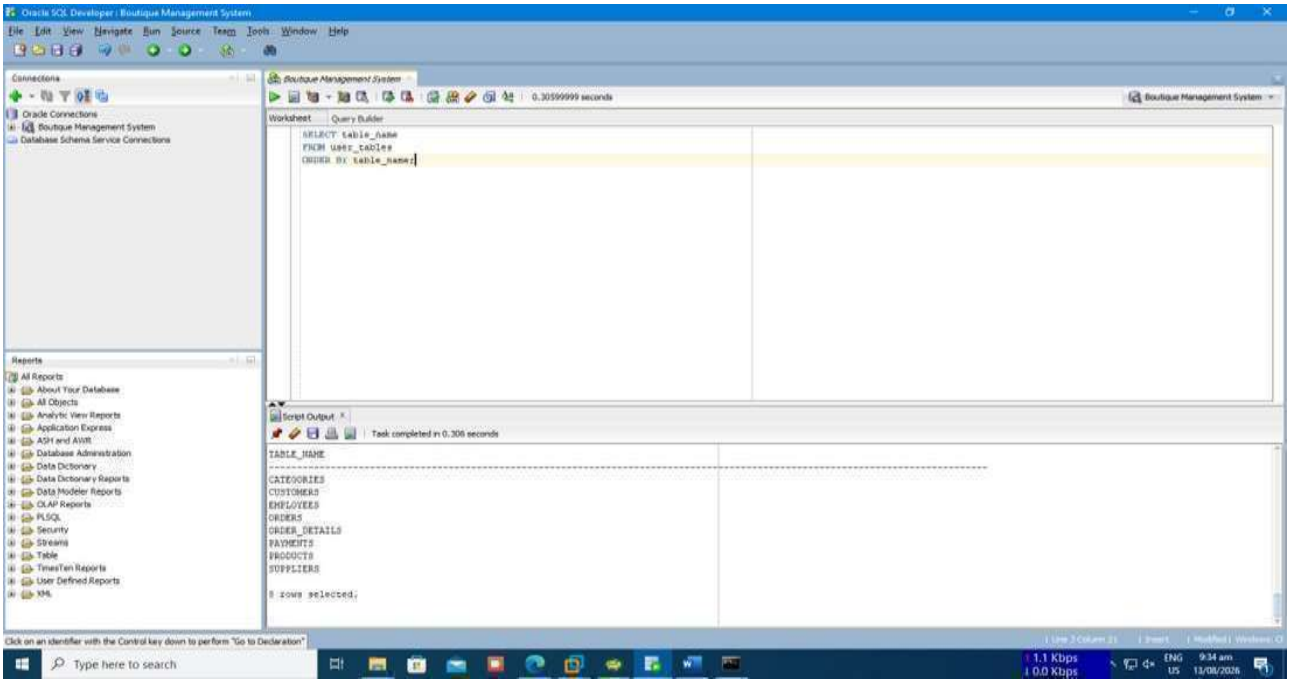


Figure 2 : Primary Key Constraints Verification

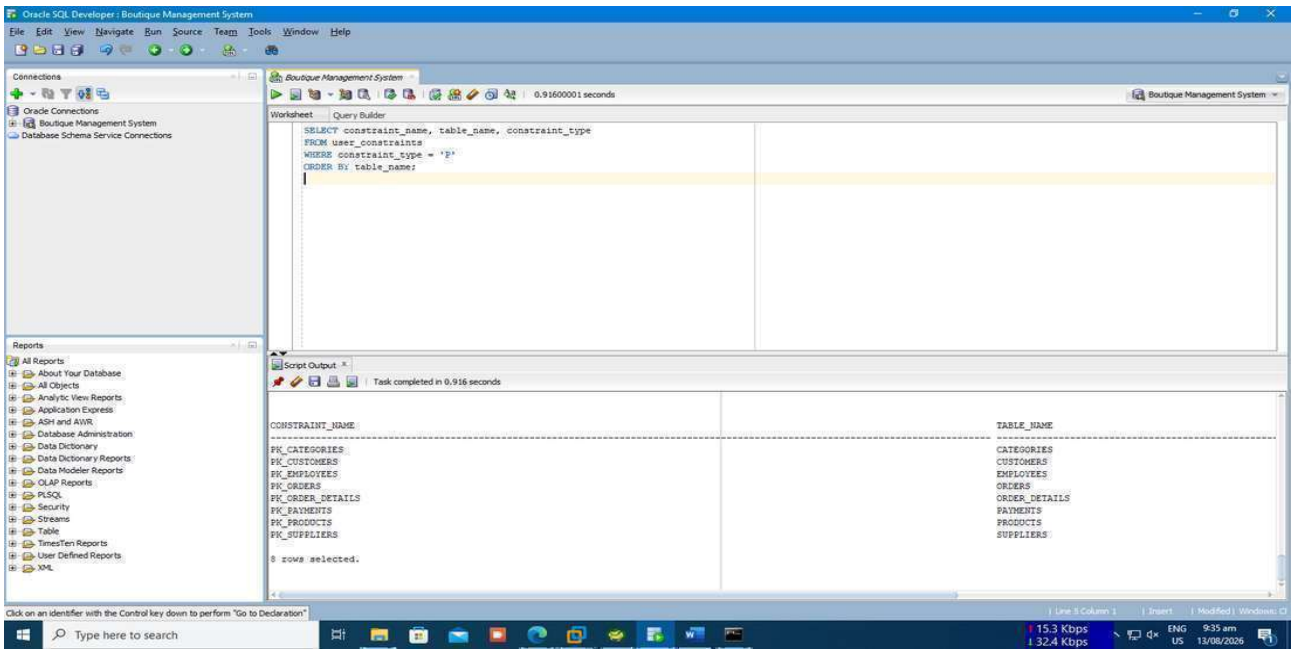


Figure 3 : Foreign Key Constraints Verification

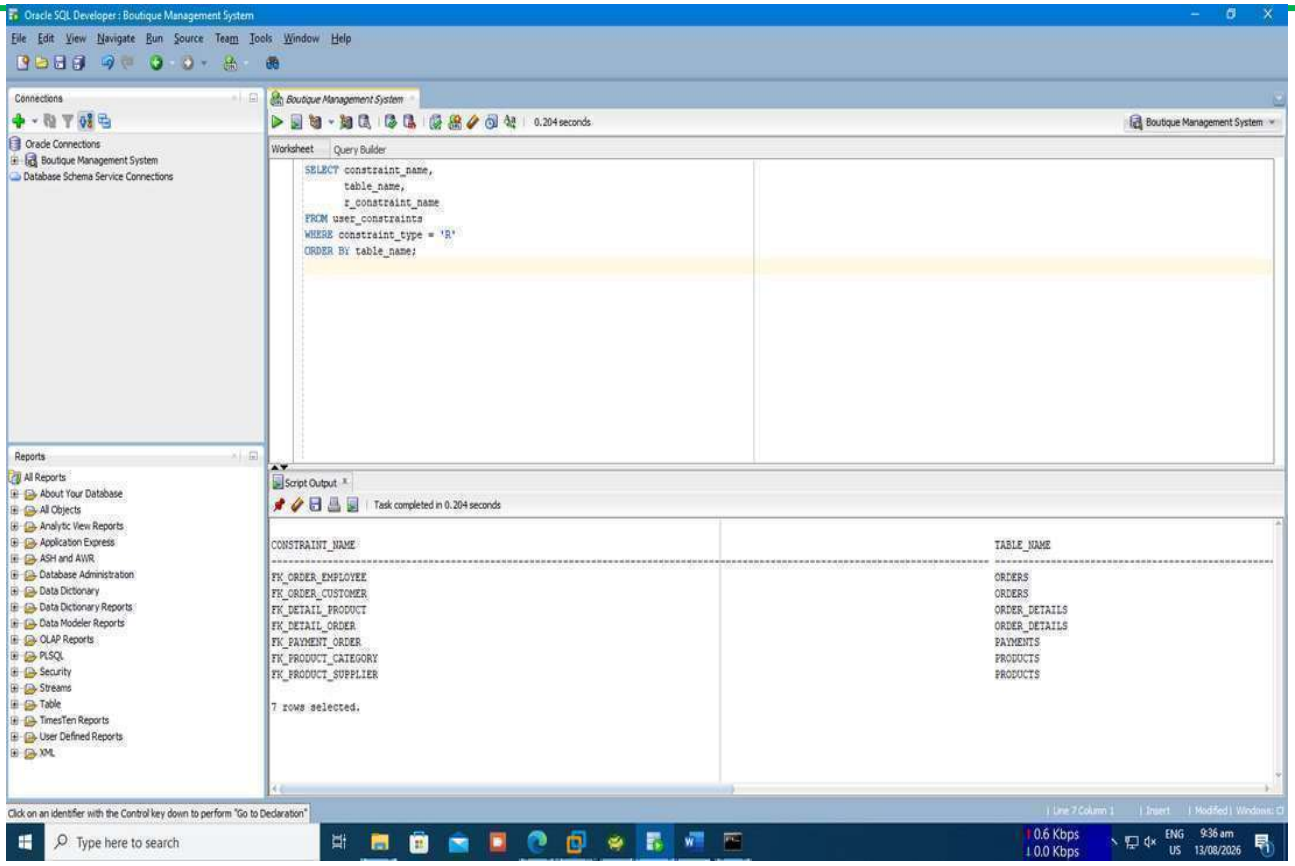


Figure 4 : UNIQUE Constraint Verification

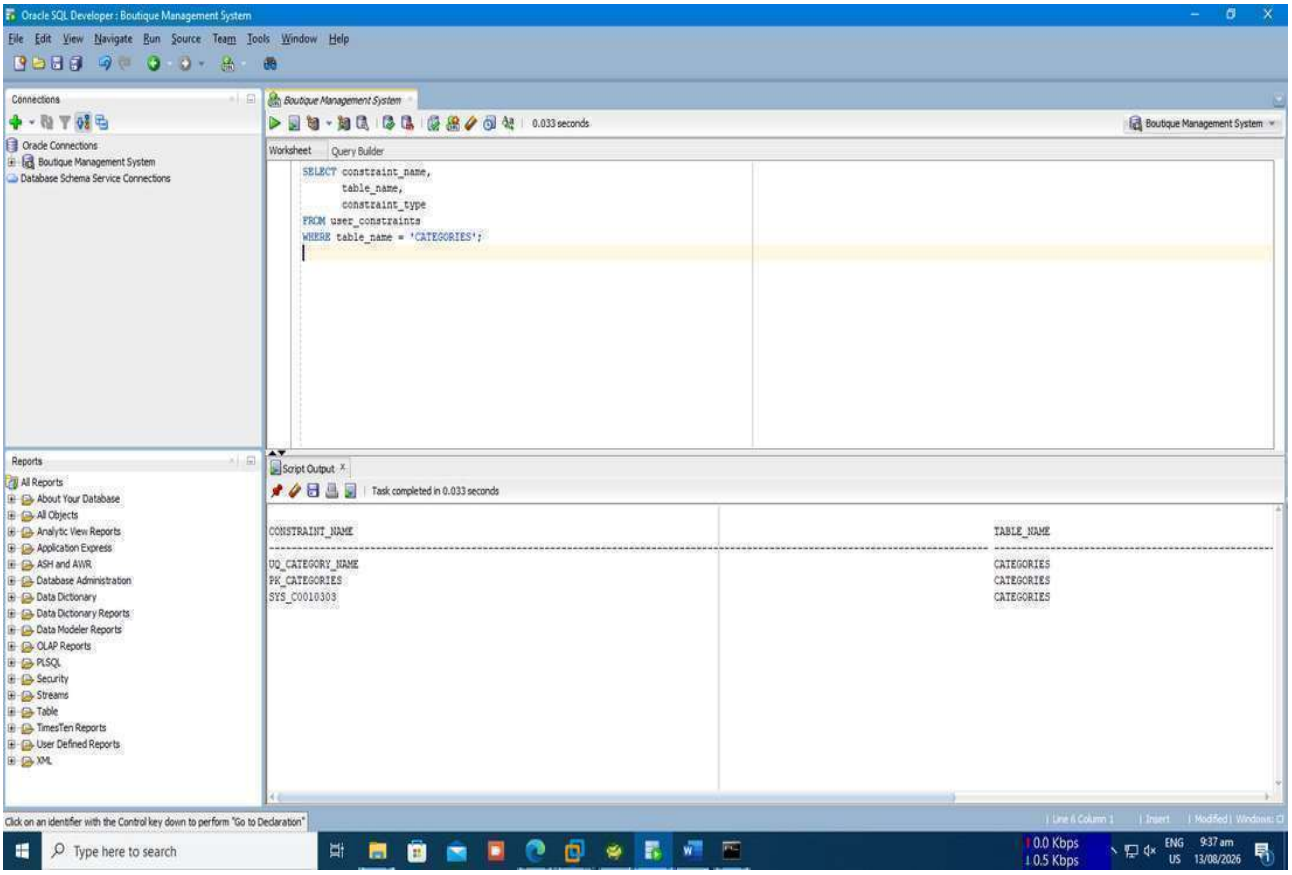


Figure 5 : NOT NULL Constraint Verification

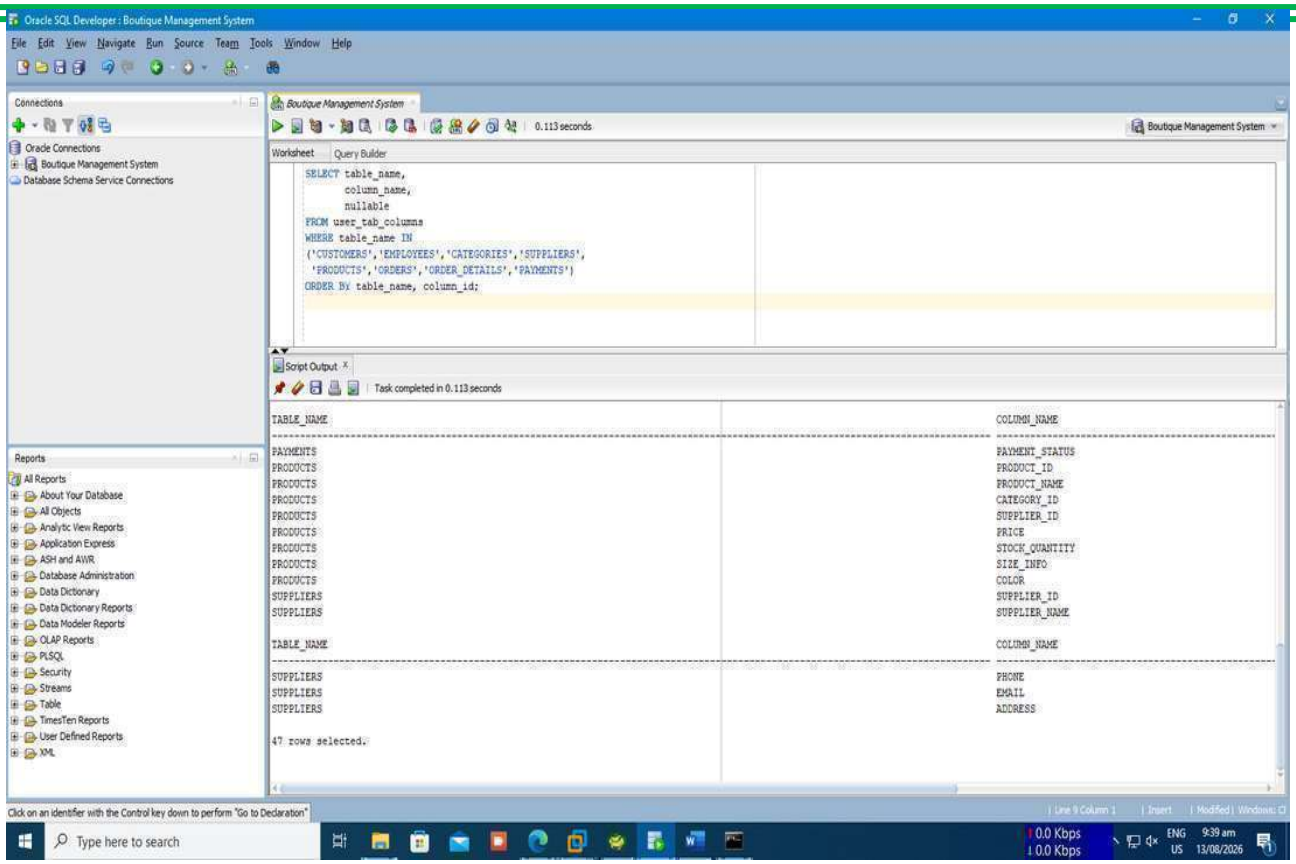


Figure 6 : CHECK Constraints Verification

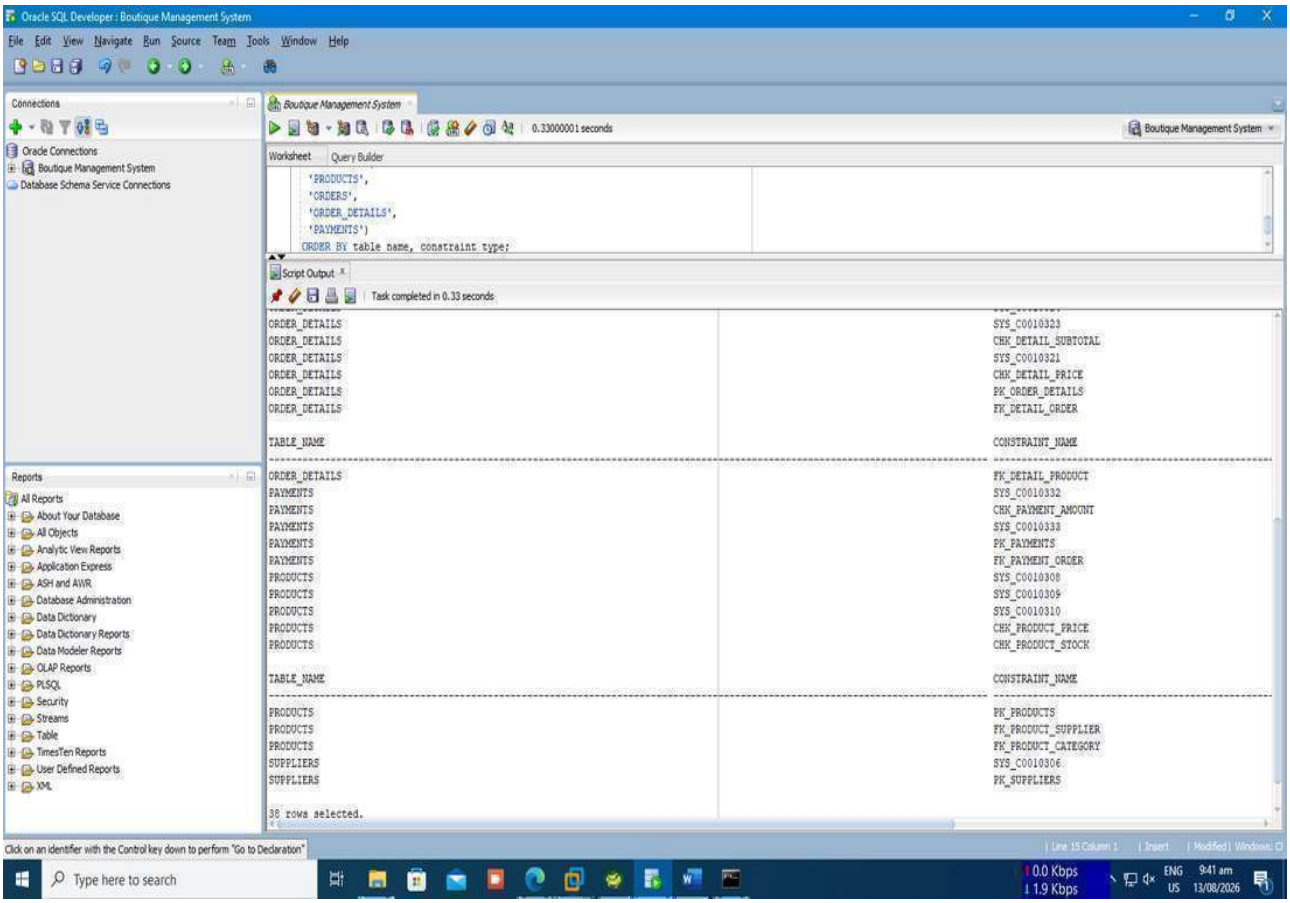


Figure 7 : Sample Data Record Count Verification

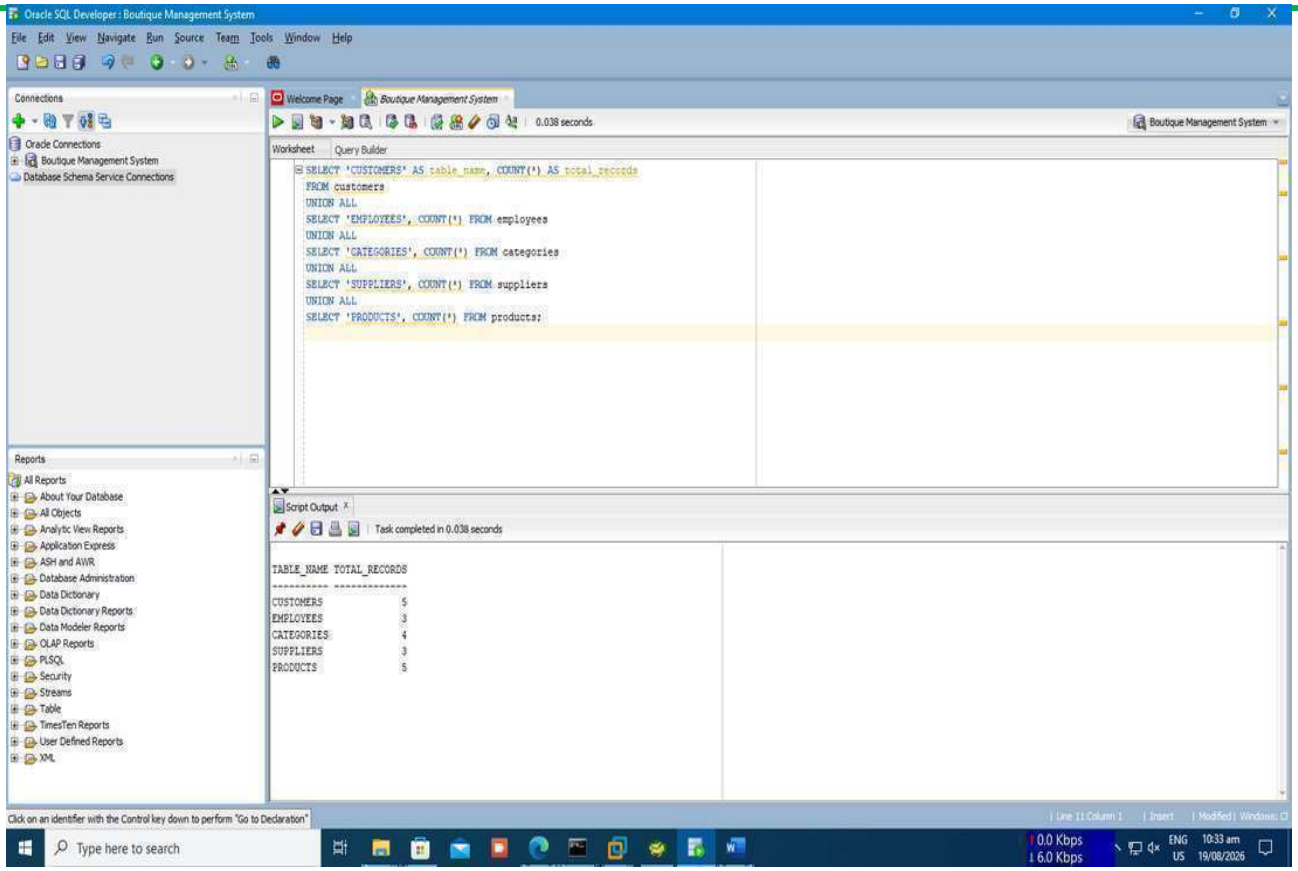
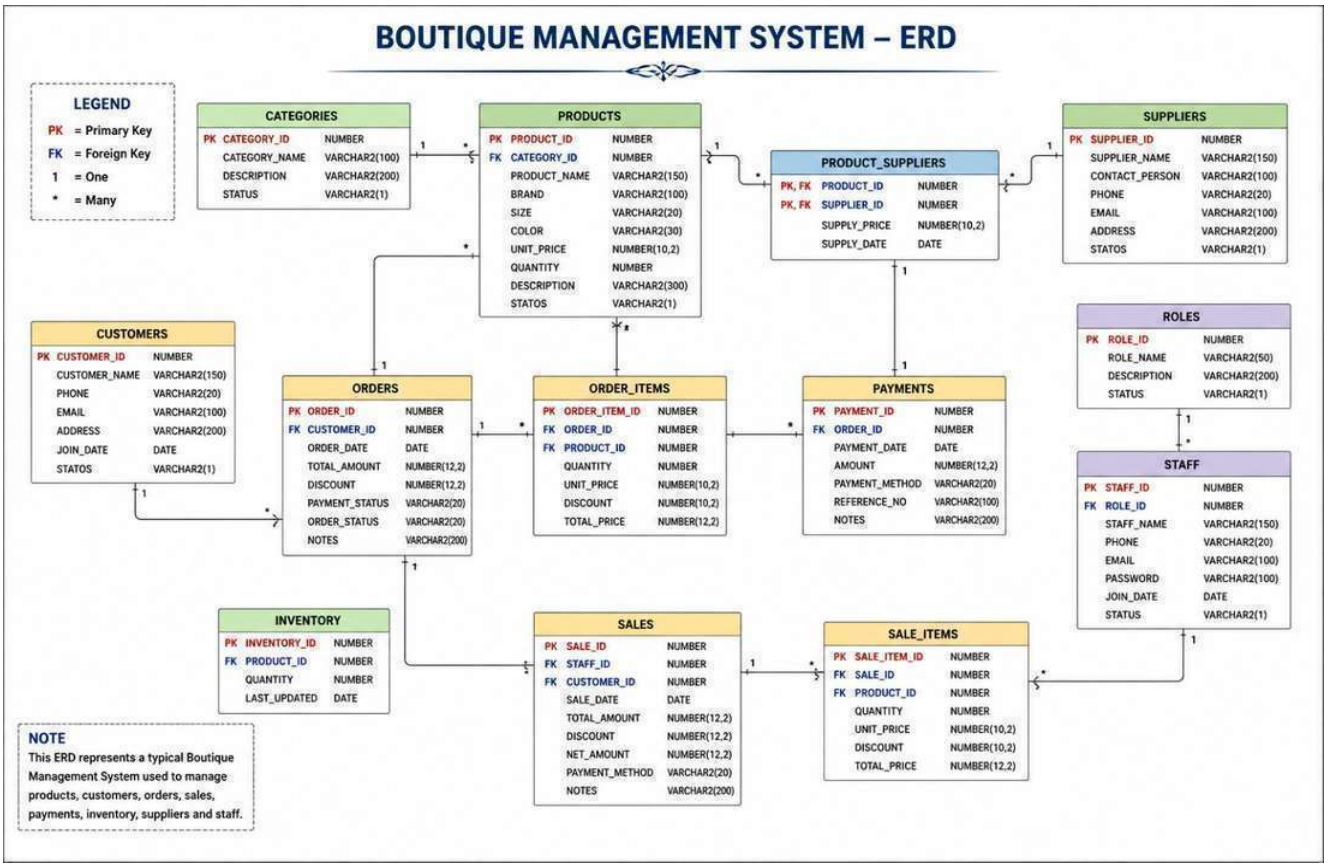


Figure 8 : Entity Relationship Diagram (ERD)



8. Week 5 : Database Administration, Backup and Data Guard

8.1 Overview

Week 5 focused on database administration through TOAD and the introduction of Oracle Enterprise Manager for database health monitoring. The emphasis was on administrative operations rather than application development.

8.2 Database Administration Through TOAD

TOAD was used to understand administrative activities such as database access, user-related operations, object management, and monitoring. The graphical interface made it easier to inspect database information and perform administrative tasks.

8.3 Remote Database Access

Remote database access was studied to understand how a DBA or authorized client can connect to an Oracle Database running on another machine. Host/IP address, listener port, service name, TNS configuration, and authentication are important components of remote connectivity.

8.4 Database Backup Concepts

Different types of database backups were discussed and practiced conceptually. Backup is a critical DBA responsibility because it provides a recovery mechanism in case of accidental deletion, hardware failure, corruption, or other operational problems.

- Full backup — protects the complete database.
- Incremental backup — captures changes since a previous backup.
- Physical backup — protects database files and physical structures.
- Logical backup — protects logical database objects and data.

8.5 Oracle Enterprise Manager

Oracle Enterprise Manager (OEM), particularly EM Express in Oracle Database 12c, was introduced as a graphical web-based monitoring and administration tool. It provides a convenient interface for viewing database availability, performance information, sessions, storage, SQL activity, and alerts.

8.6 Database Health Monitoring

Database health was examined through the monitoring interface. Important areas included database availability, performance, sessions, storage/tablespaces, SQL activity, and alerts. These indicators help a DBA identify potential problems before they affect users.

Figure 1 : Screenshot 7.2: Creating Database Link

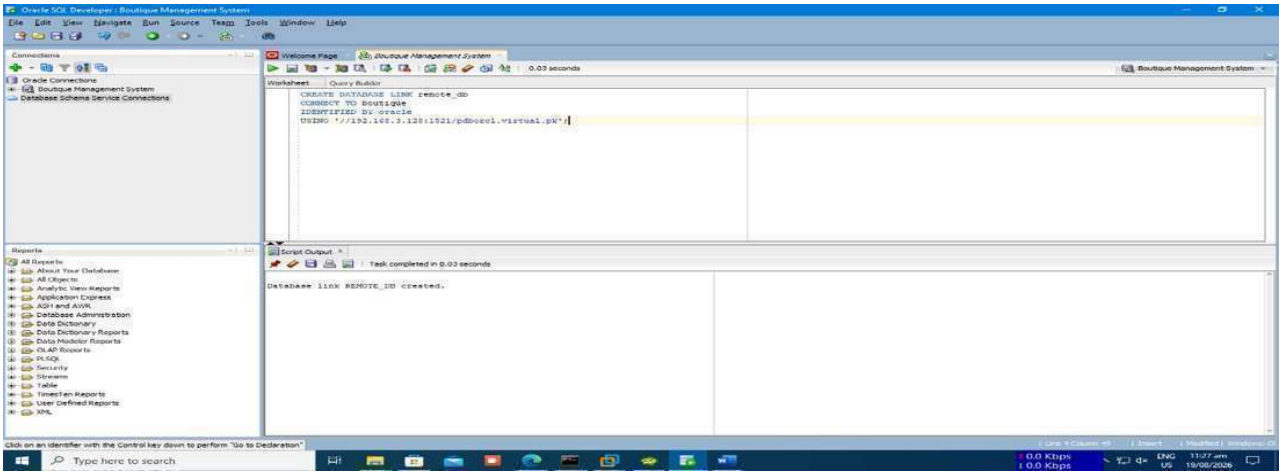


Figure 2 : Database Link Verification

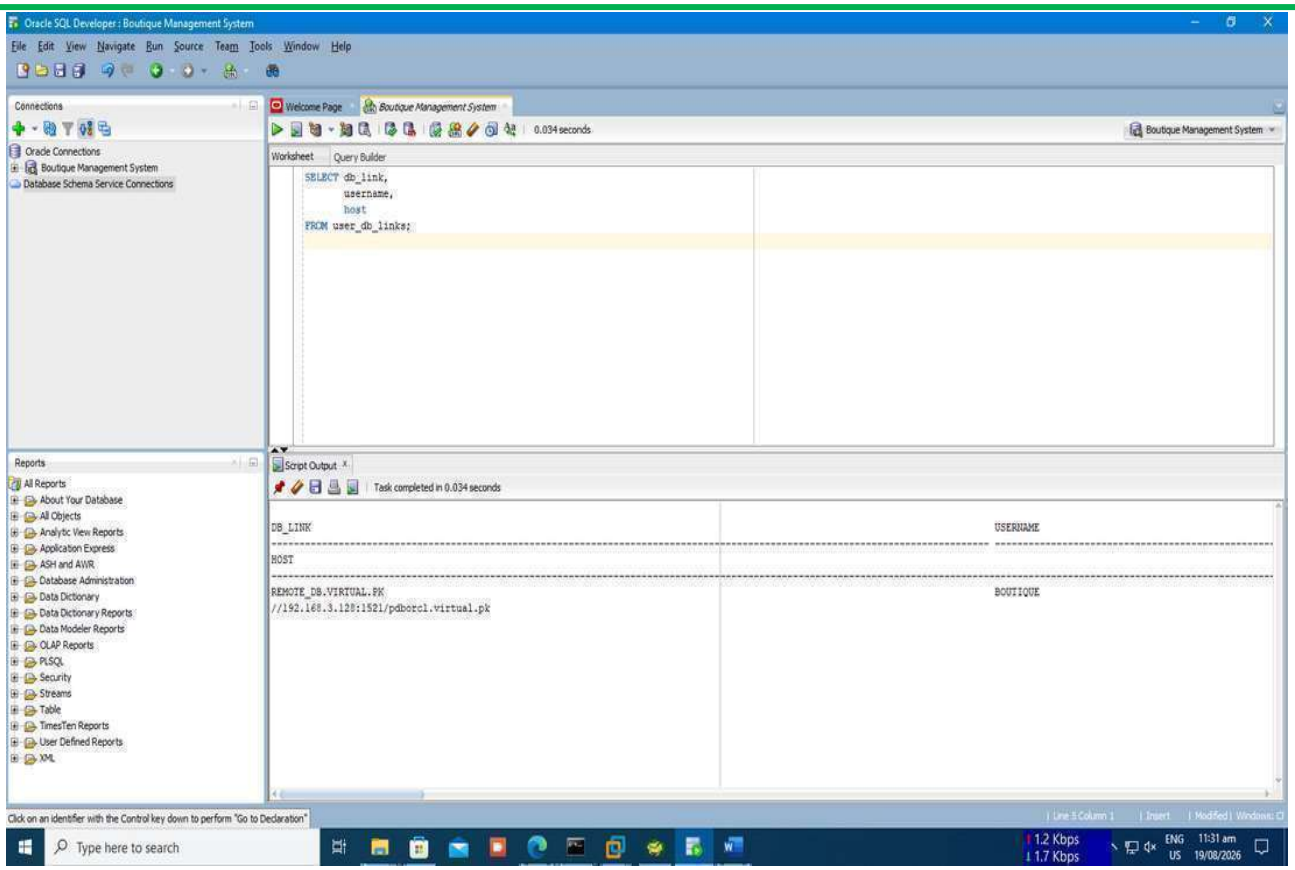


Figure 3 : Connecting to the Oracle Database Using RMAN

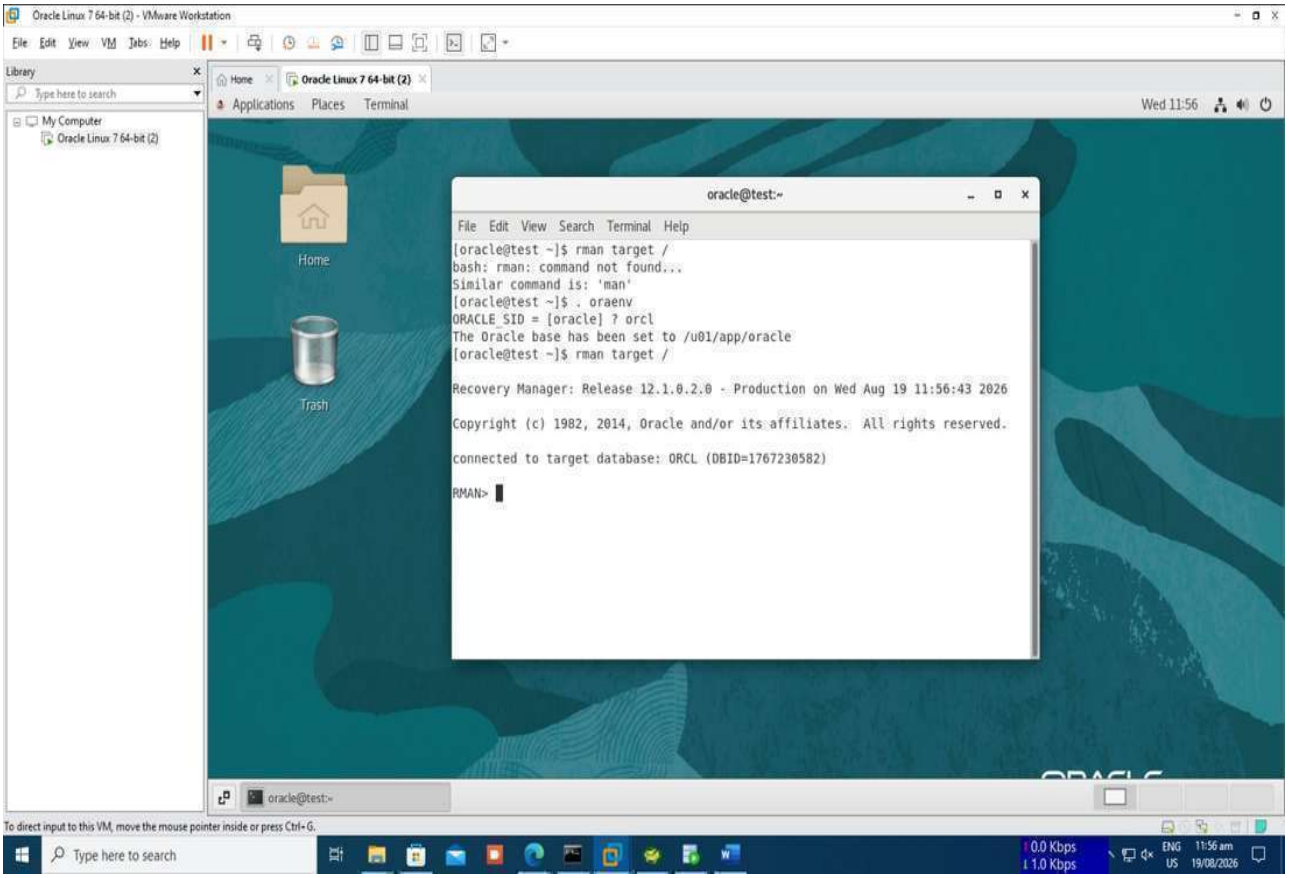


Figure 4 : Starting the Database in MOUNT Mode

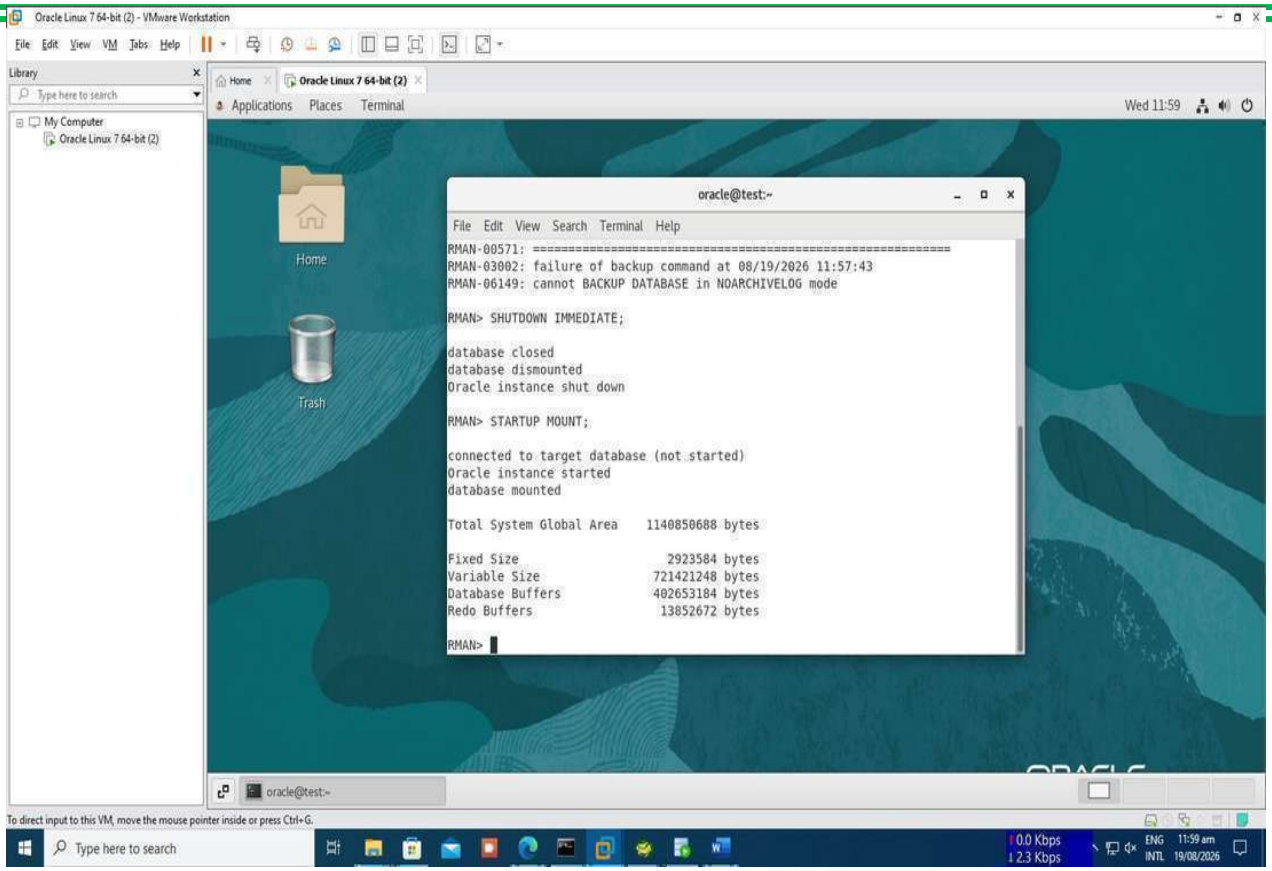


Figure 5 : Successful RMAN Database Backup

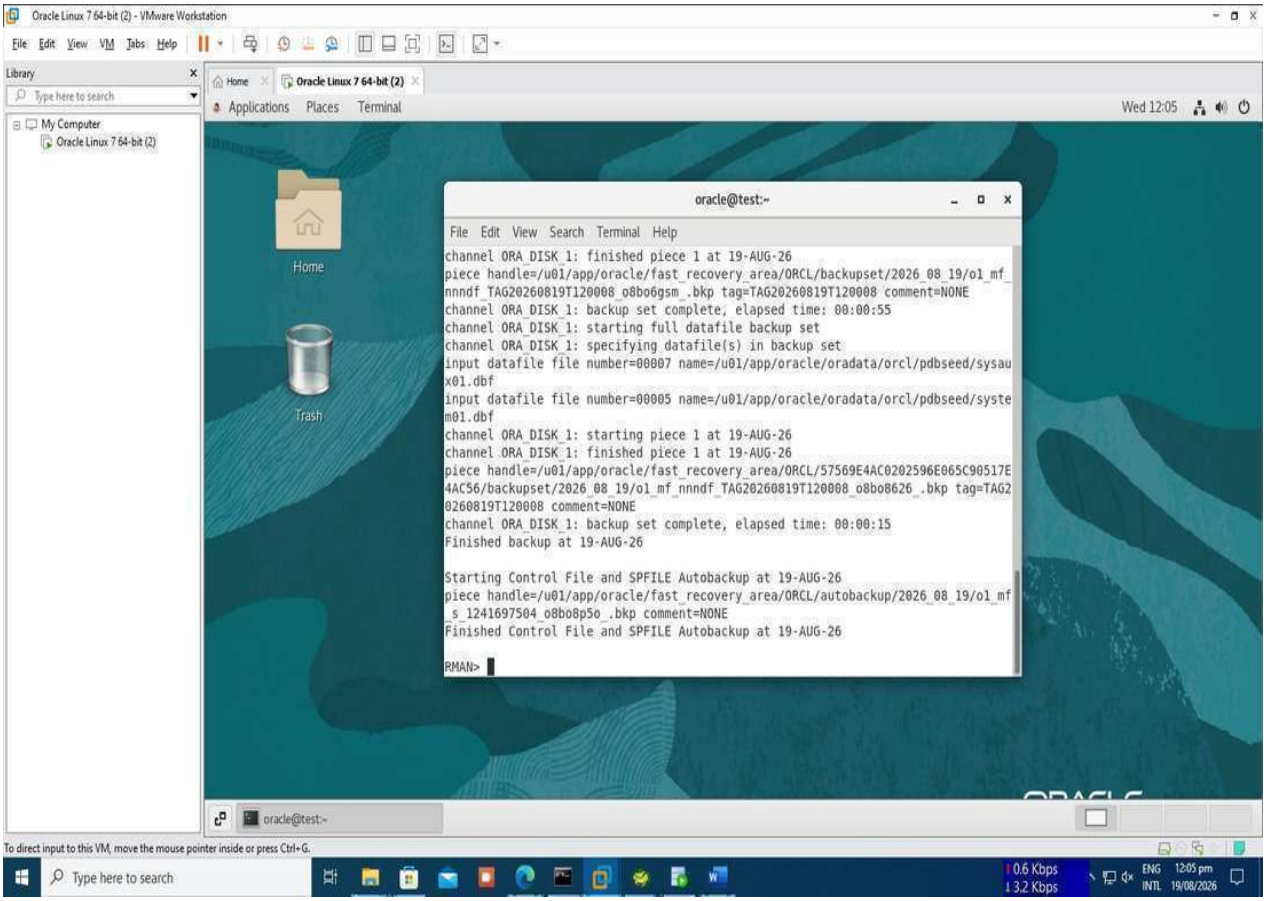


Figure 6 : Opening and Verifying the Database

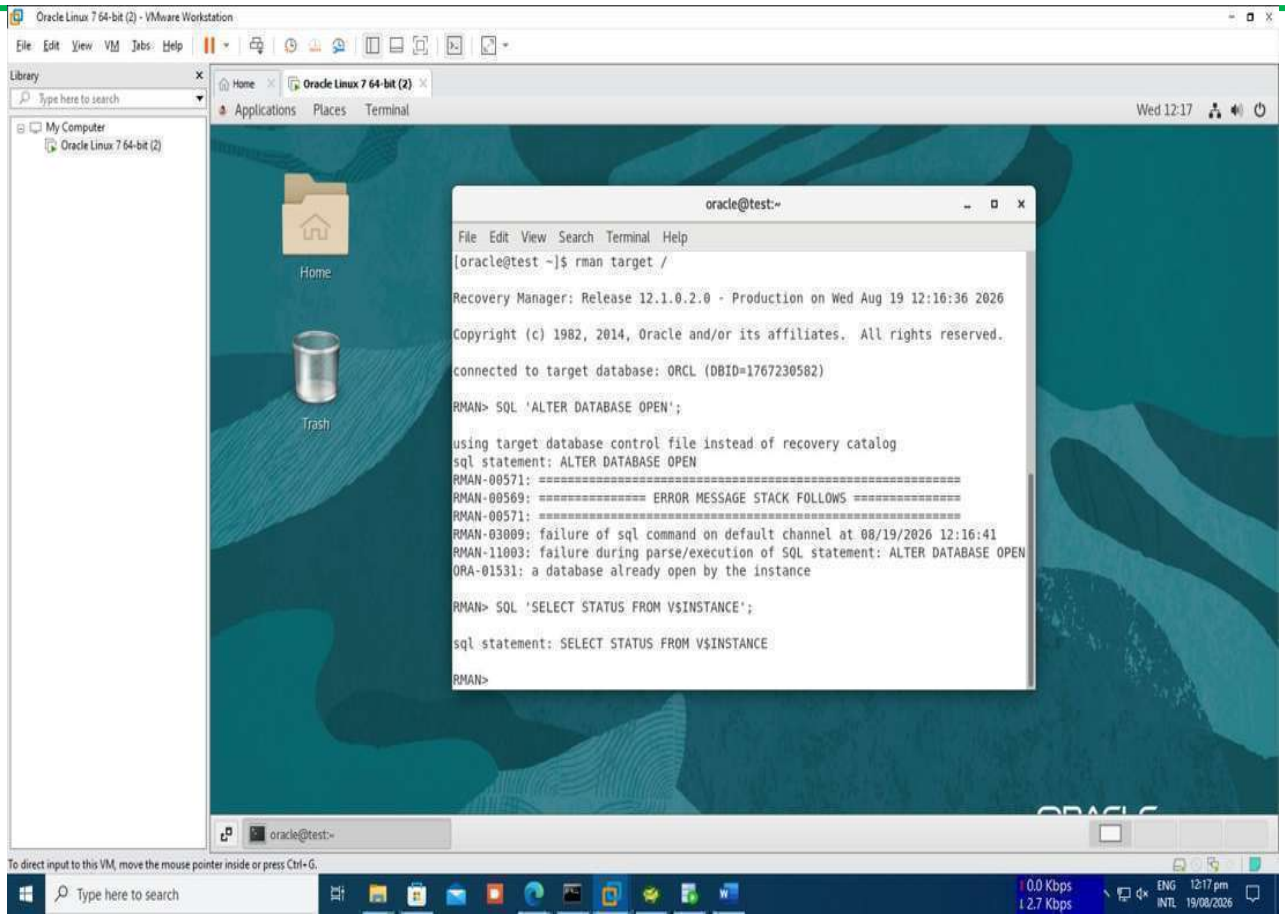


Figure 7 : RMAN Backup Verification

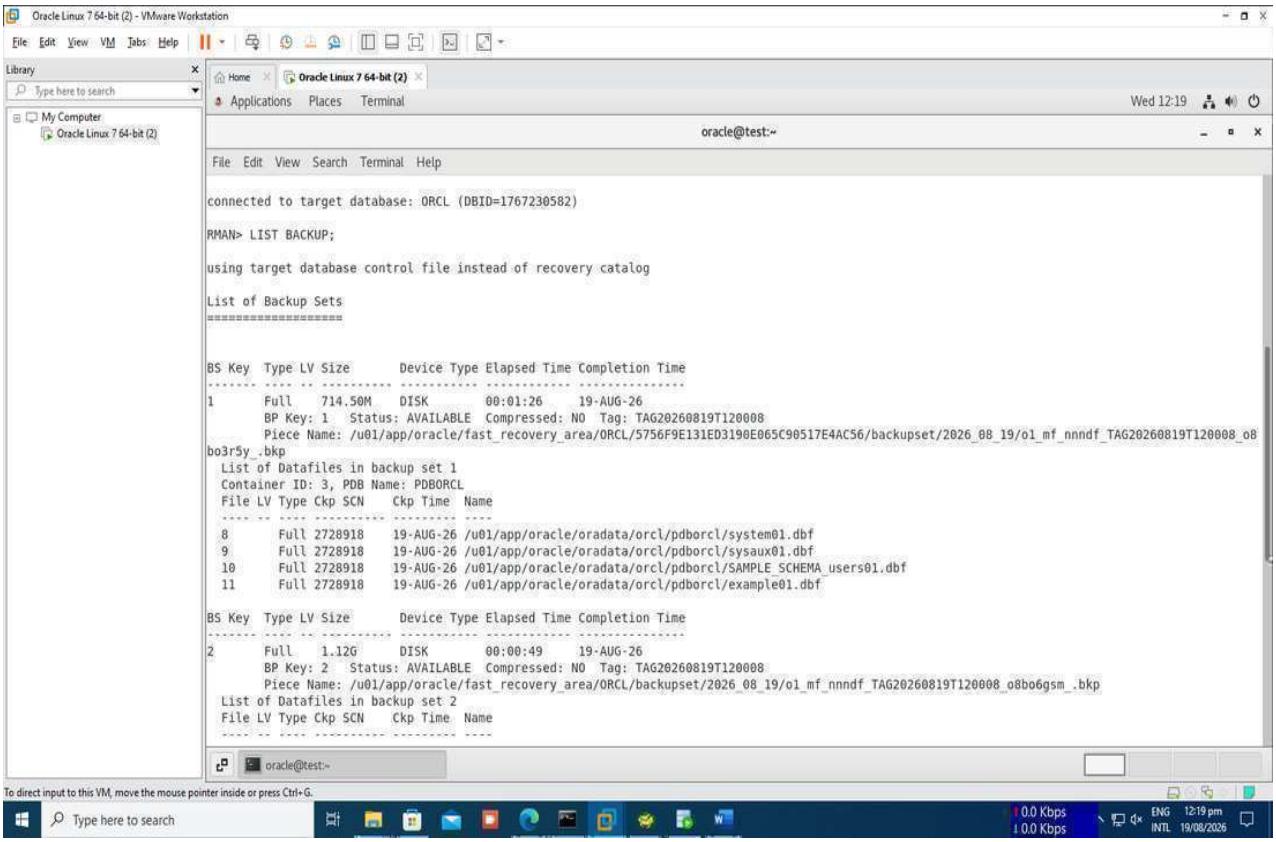
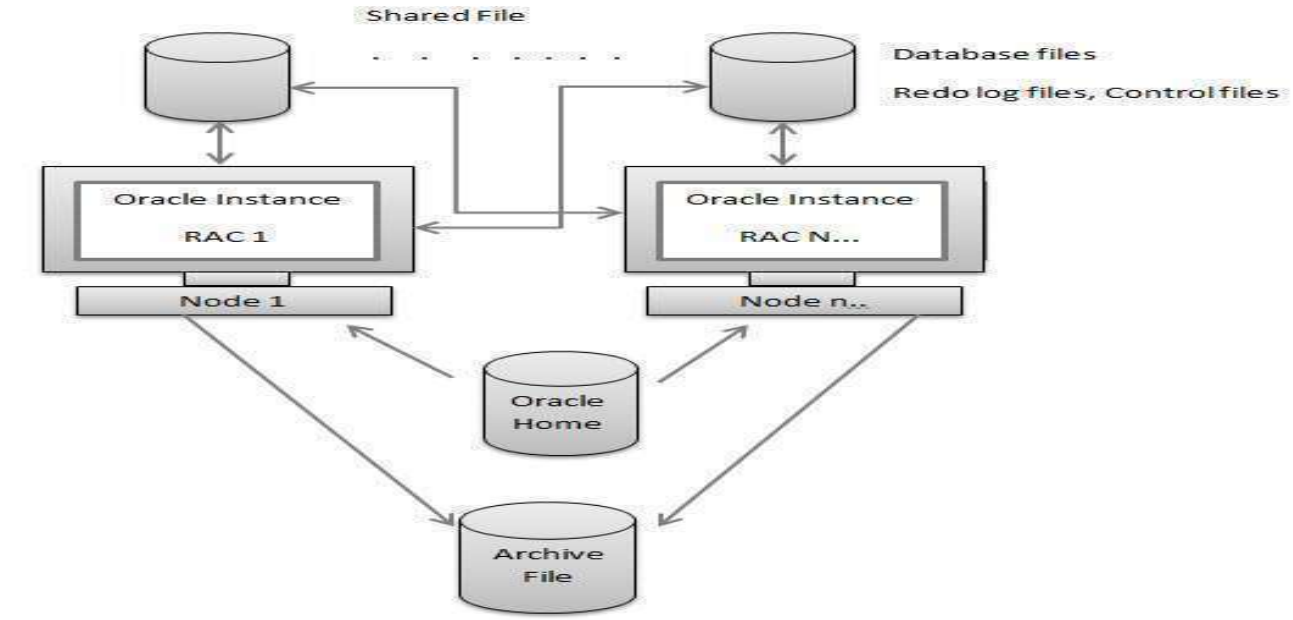


Figure 8 : Oracle RAC Architecture



RAC Architecture

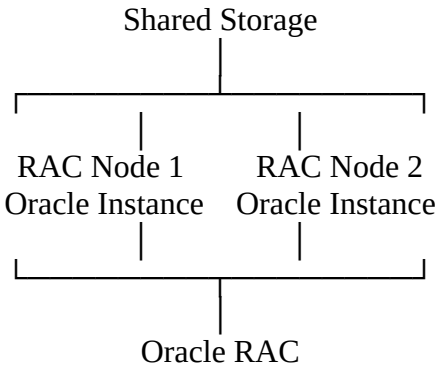


Figure 9 : Database Session Monitoring

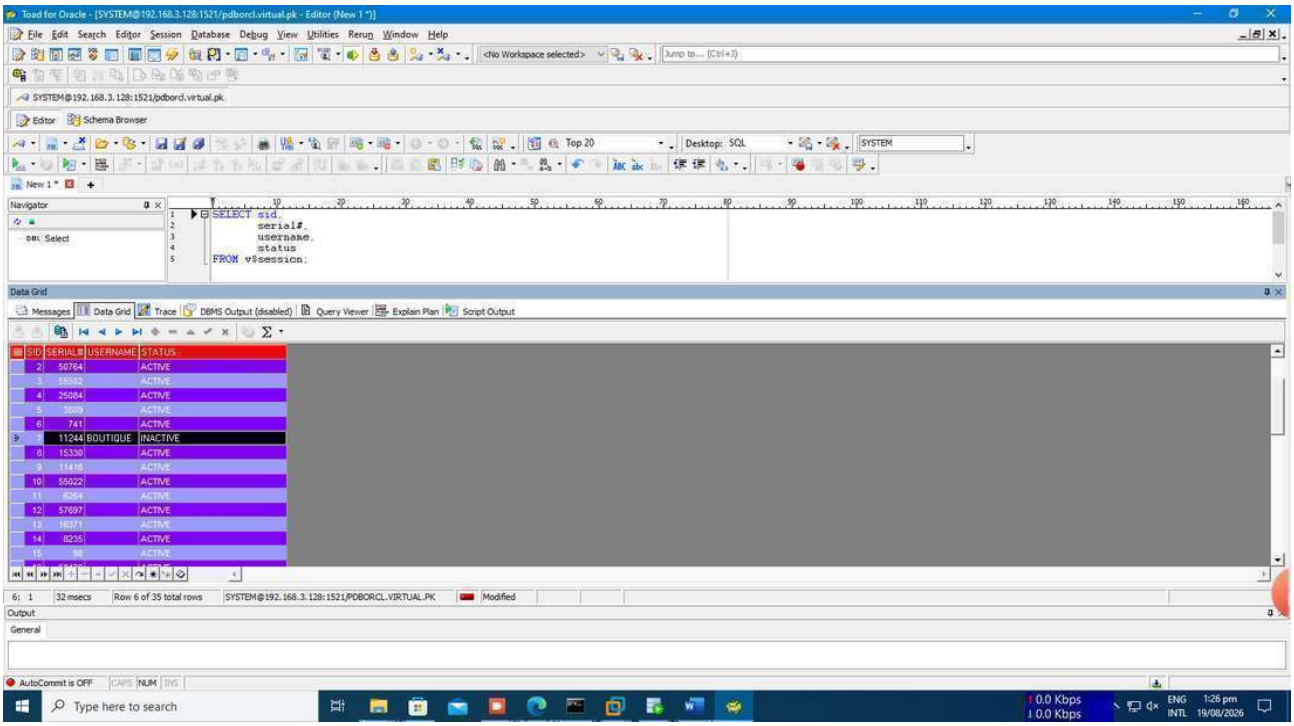


Figure 10 : Database Status

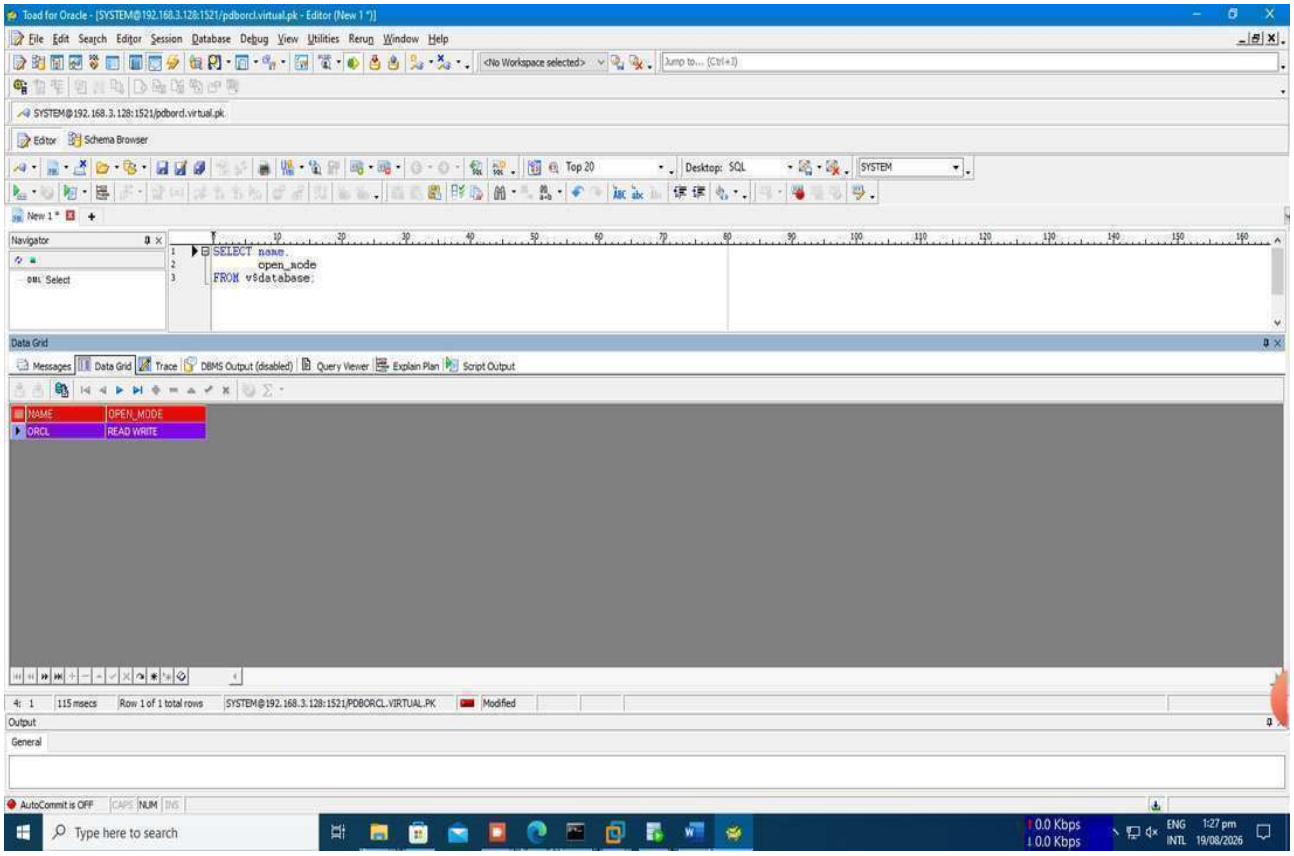


Figure 11 : Tablespace Monitoring

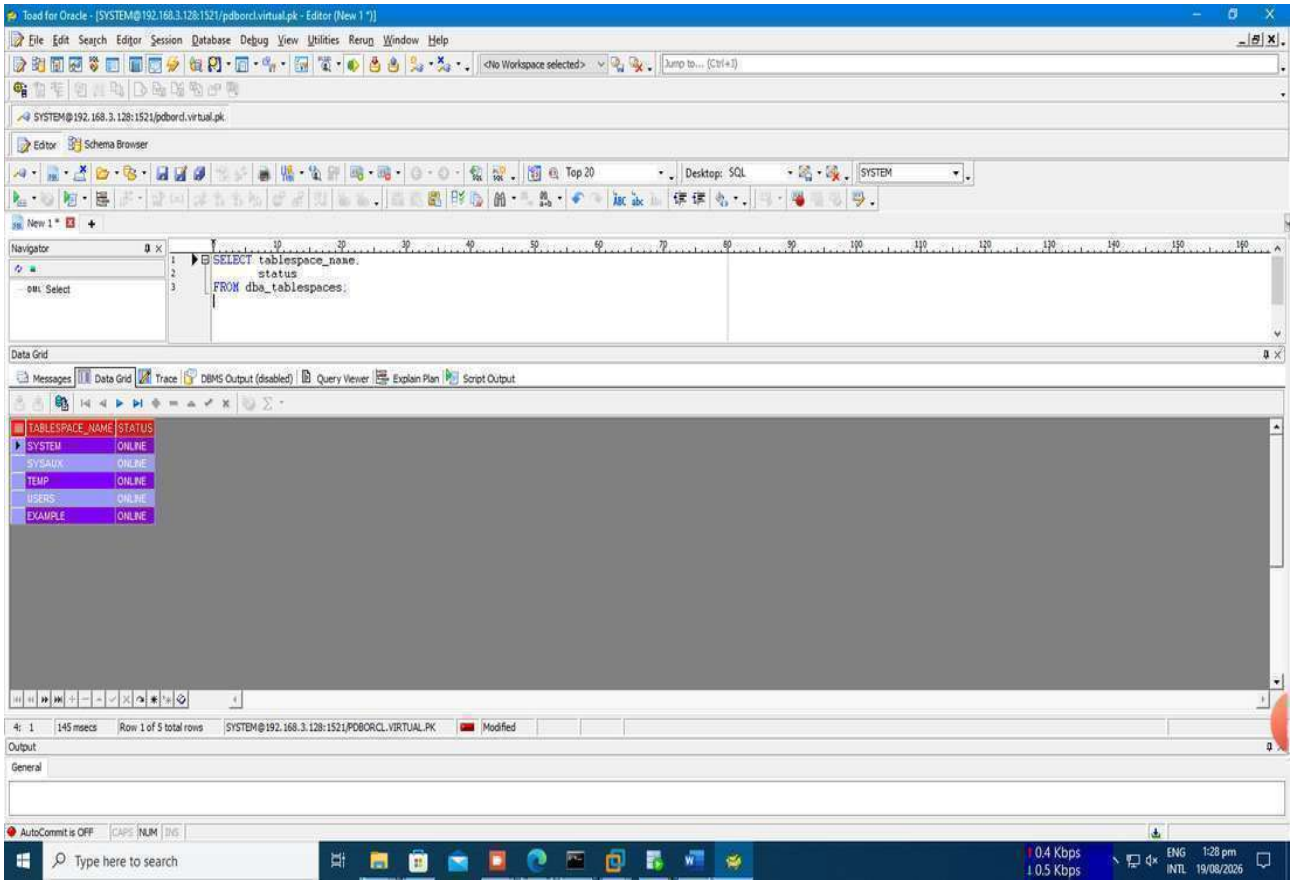
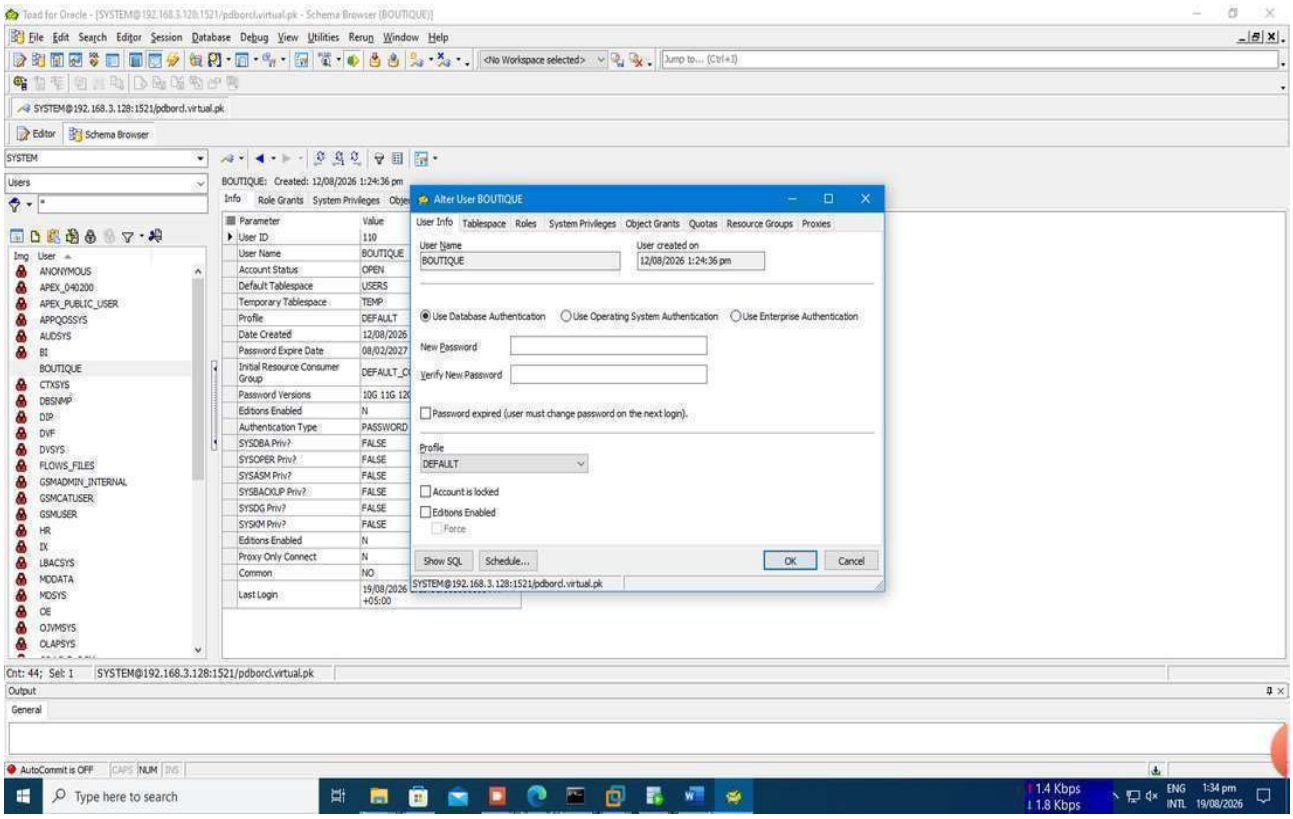


Figure 12 : Database Security and User Management Environment



8.7. Oracle Data Guard

Oracle Data Guard is an Oracle Database feature used to provide **high availability, data protection, and disaster recovery**. It maintains one or more standby databases for a primary database. The changes made in the primary database are transferred to the standby database through redo data.

The primary database is the main database where normal database operations are performed. A standby database acts as a backup and can be used if the primary database becomes unavailable.

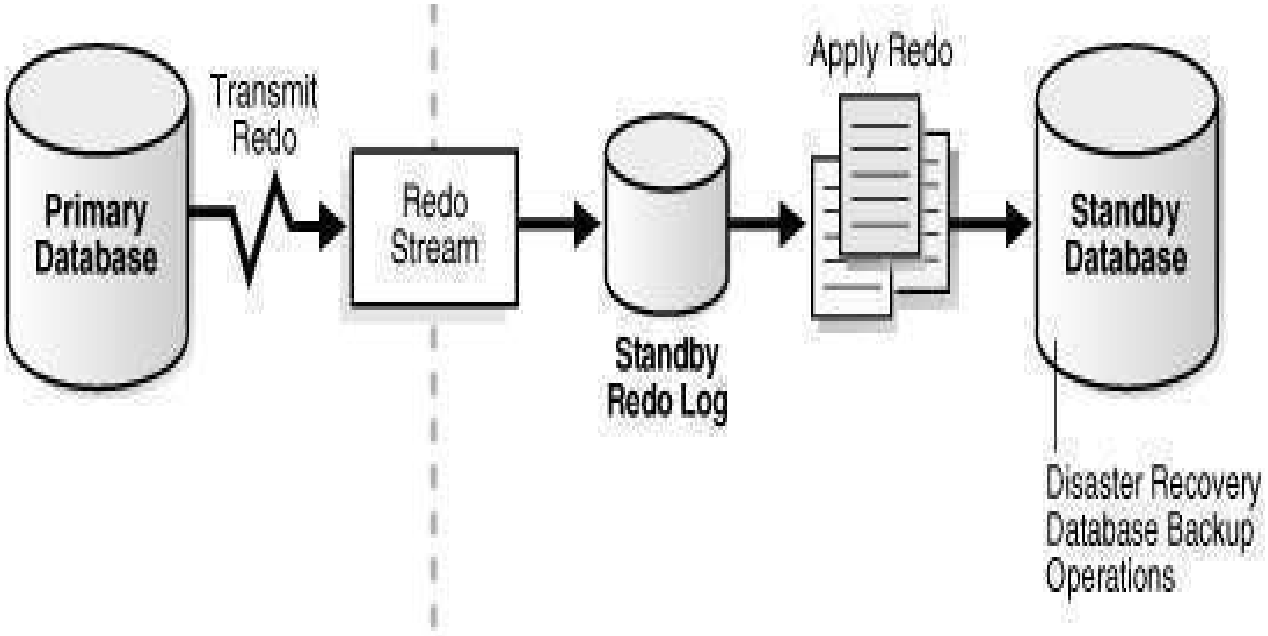
Main Components

- **Primary Database:** The main production database.
- **Standby Database:** A copy of the primary database maintained for disaster recovery.
- **Redo Data:** Contains information about changes made in the database.
- **Data Guard Services:** Manage the transfer and application of redo data between primary and standby databases.

Benefits of Data Guard

- Provides disaster recovery.
- Improves database availability.
- Protects against data loss.
- Supports planned and unplanned database failover.
- Helps maintain a standby copy of the production database.

Figure 13 : Oracle Data Guard Primary and Standby Database Architecture



8.8. Synonyms

A synonym is an alternative name or alias for a database object such as a **table, view, sequence, procedure, or another database object**. Synonyms make it easier for users and applications to access database objects without specifying the complete object name or owner.

For example, if a table belongs to another schema, a synonym can be created so that the user can access the table using a simpler name.

Types of Synonyms

There are mainly two types of synonyms:

1. **Private Synonym** – Available only to the user who creates it.
2. **Public Synonym** – Available to all database users who have the required privileges.

Example

CREATE SYNONYM emp_syn

FOR hr.employees;

After creating the synonym, the user can access the object using:

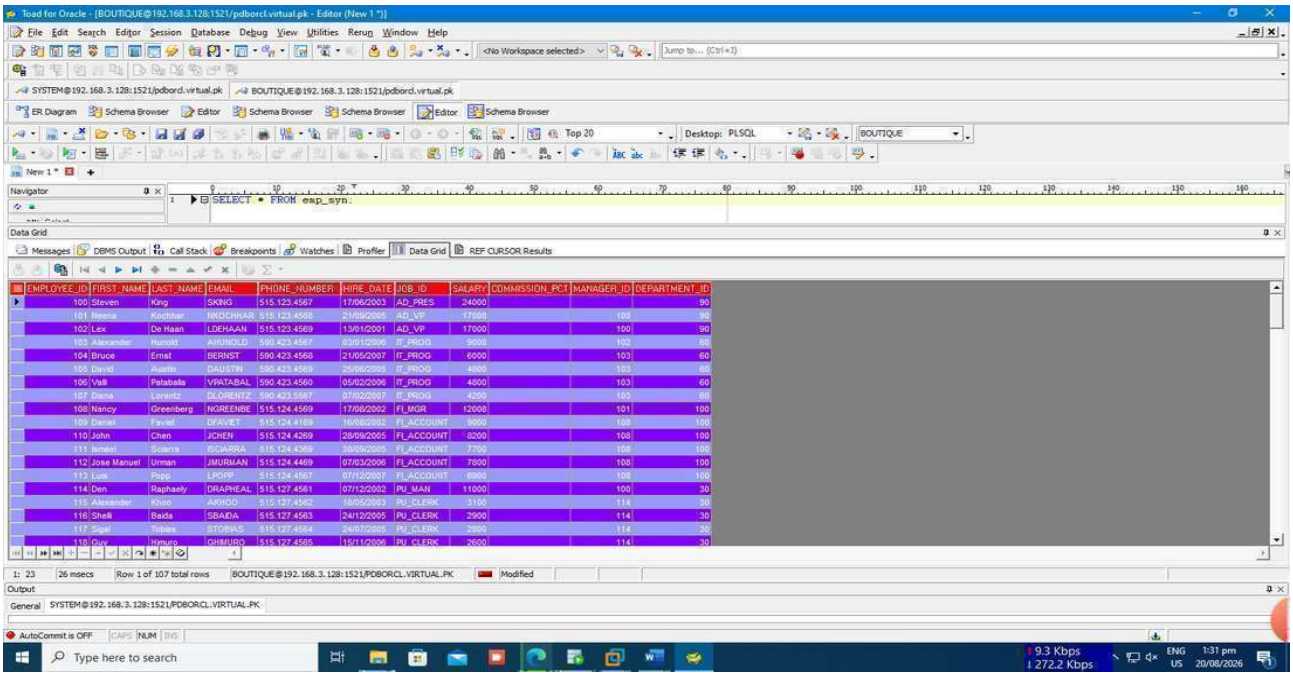
SELECT * FROM emp_syn;

Instead of:

SELECT * FROM hr.employees;

Synonyms are useful for simplifying SQL statements, hiding the owner/schema name, and providing a convenient way to access database objects.

Figure 14 : Synonym Creation and Execution



8.9. Database Auditing

Database auditing is the process of **monitoring and recording activities performed within a database**. It helps database administrators identify who accessed the database, what actions were performed, and when those actions occurred.

Auditing is an important part of database security because it provides a record of user activities and helps detect unauthorized or suspicious operations.

Common Activities That Can Be Audited

- User logins and logouts.
- Data modification activities.
- INSERT, UPDATE, and DELETE operations.
- CREATE, ALTER, and DROP operations.
- Access to important database objects.
- Security-related activities.

Importance of Auditing

Database auditing helps in:

- Improving database security.
- Monitoring user activities.
- Detecting unauthorized access.
- Maintaining accountability.
- Supporting troubleshooting and investigation.
- Meeting security and compliance requirements.

A DBA can configure auditing according to the organization's security requirements and then review the generated audit information.

Figure 15 : Database Auditing Configuration

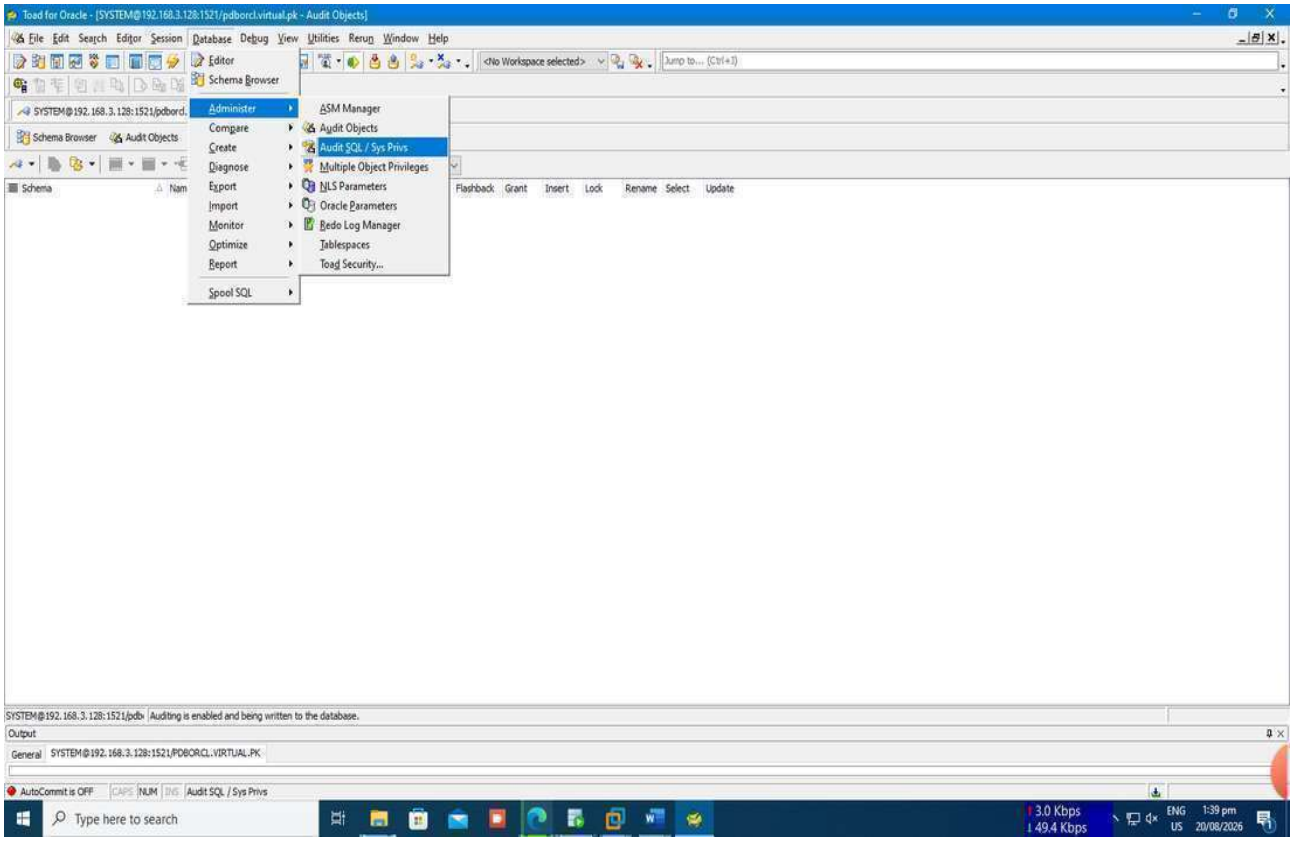
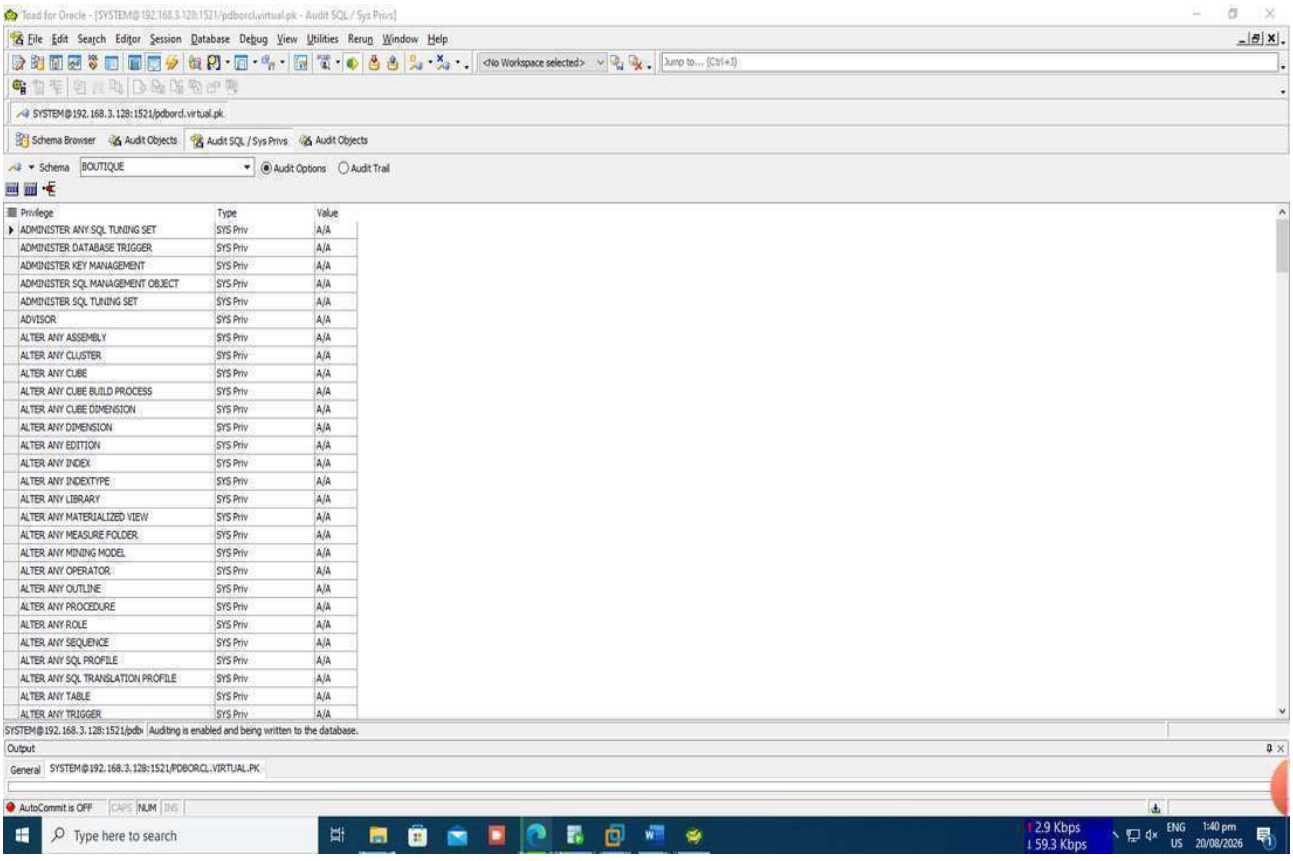


Figure 16 : Viewing Database Audit Records



9. Week 6 : Data Center Visit and DBA Challenges

9.1 Overview

The final week provided exposure to a real Data Center environment and summarized the technical challenges encountered during the internship.

9.2 Data Center Visit

The instructor arranged a Data Center visit to provide practical exposure to real IT infrastructure. Different types of servers were observed, and their purpose and working were explained.

9.3 Oracle Database Server

An Oracle Database Server was observed during the visit. Its role in centralized data storage, database processing, application support, user access, and database services was discussed.

9.4 Data Center Infrastructure

The visit also provided awareness of server racks, networking, storage, backup systems, power arrangements, cooling, and physical security. These supporting components are essential for reliable IT services.

Figure 1 : Data Center Overview



Figure 2 : Servers Installed in the Data Center



Figure 3 : Oracle Database Server



Figure 4 : Server Rack and Data Center Infrastructure



9.5 Internship Challenges

Throughout the internship, practical problems were encountered in Oracle Linux, database installation, listener configuration, TNS connectivity, SQL Developer, TOAD, PDB services, authentication, and network communication. These problems provided practical troubleshooting experience.

9.6 Major Errors and Challenges

Issue / Error	General Solution
sqlplus: command not found	Configured Oracle environment variables and PATH.
ORA-12514	Checked listener services and database service registration.
ORA-12154	Checked TNS alias and tnsnames.ora configuration.
ORA-12541	Verified and started the Oracle Listener.
ORA-01017	Verified database credentials.
PDB service registration issue	Checked PDB status and listener service registration.
Network connectivity problem	Checked IP address, interface, port, and network configuration.
TOAD / SQL Developer connection failure	Verified host, port, service name, listener, TNS, and credentials.

9.7 Main Learning

Week 6 demonstrated the relationship between DBA knowledge and real infrastructure. The Data Center visit provided professional exposure, while the troubleshooting summary demonstrated the importance of systematic problem solving.

10. Overall Learning Outcomes

The six-week internship provided a progressive learning path from infrastructure preparation to Oracle Database administration and troubleshooting.

Area	Learning Achieved
Virtualization	Prepared an Oracle Linux virtual machine using VMware.
Operating System	Worked with Oracle Linux for Oracle Database administration.
Database Installation	Installed and configured Oracle Database 12c.
Connectivity	Connected SQL Developer and TOAD with Oracle Database.
Database Design	Designed and implemented the Boutique Management System.
Administration	Performed basic DBA tasks through TOAD.
Remote Access	Understood network-based database connectivity.
Backup	Studied important database backup concepts.
Monitoring	Used OEM/EM Express concepts to examine database health.
Troubleshooting	Analyzed Oracle, TNS, listener, Linux, and network problems.
Infrastructure	Observed a professional Data Center environment.

10.1 Technical Knowledge

The internship strengthened knowledge of Oracle Database architecture, Linux-based database environments, listener services, TNS connectivity, client tools, database design, user administration, backup concepts, monitoring and troubleshooting.

10.2 Practical Confidence

Repeated implementation and troubleshooting improved confidence in using command-line utilities, graphical database tools, and systematic diagnostic procedures.

11. Professional Skills Developed

- Database administration and monitoring
- Oracle Linux command-line usage
- Oracle Listener and TNS troubleshooting
- SQL Developer and TOAD usage
- Database design and relational modeling

- User and privilege management
- Backup and recovery awareness
- Problem identification and troubleshooting
- Technical documentation
- Professional communication and teamwork
- Data Center and enterprise infrastructure awareness

11.1 Problem-Solving Approach

A key professional skill developed during the internship was the ability to approach database problems systematically. Instead of treating an error message as an isolated failure, the relevant layer was identified first—client, network, listener, TNS, authentication, database service, or operating system—followed by verification and correction.

11.2 Documentation Skills

The internship also improved technical documentation skills by recording procedures, commands, outputs, screenshots, errors, solutions, and learning outcomes in a structured format.

12. Conclusion

The six-week internship provided a comprehensive practical introduction to Database Administration with a focus on Oracle Database 12c. The training started with VMware and Oracle Linux preparation, followed by Oracle Database installation and configuration.

The next stage focused on SQL Developer and TOAD connectivity, which provided practical understanding of database client tools and network-based database access. The Boutique Management System project then applied relational database design concepts in a practical business scenario.

Database administration activities in Week 5 expanded the learning toward remote access, backup concepts, TOAD administration, and Oracle Enterprise Manager monitoring. The final week provided real-world exposure through a Data Center visit and consolidated the troubleshooting experience gained throughout the internship.

Overall, the internship successfully connected academic database concepts with practical DBA activities. It improved technical knowledge, troubleshooting ability, documentation skills, and awareness of professional database infrastructure.

13. Recommendations

Based on the internship experience, the following recommendations are proposed for future training:

- Provide additional hands-on sessions for Oracle backup and recovery.
- Include more practical exercises on RMAN and recovery scenarios.
- Provide additional Data Center visits to observe enterprise database infrastructure.
- Include performance tuning exercises using realistic workloads.
- Provide more practice with Oracle Enterprise Manager and monitoring dashboards.
- Include additional exercises on high availability and disaster recovery.
- Continue using real troubleshooting scenarios because they significantly improve DBA skills.

14. References

- Oracle Database 12c Documentation — Oracle Database Concepts and Administration.
- Oracle SQL Developer Documentation — Oracle Corporation.
- Oracle Enterprise Manager / Enterprise Manager Database Express Documentation.
- Oracle Database SQL Language Reference.
- Oracle Linux Documentation.
- VMware Workstation Documentation.
- TOAD for Oracle Product Documentation.

Final Internship Summary

This report summarizes the six-week practical internship in Database Administration. The training progressed from virtualization and operating system setup to Oracle Database installation, client connectivity, database development, administration, monitoring, Data Center observation and troubleshooting.

The internship provided a strong foundation for further study and professional development in Oracle Database Administration and related IT infrastructure.

Student Signature: _____

Site Supervisor Signature: _____

Date: _____